THE EYE

VOLUME 3

Technical Architecture

The authoritative logical and operational architecture of The Eye

OBSERVE / UNDERSTAND / PREDICT / SIMULATE / DECIDE / LEARN

CONTROLLED TECHNICAL ARCHITECTURE BASELINE / V1.0

Inherited from Volume 0 — Product Constitution and Volume 1 — Executive Vision Book

1 August 2026

Document Control

Volume 3 defines the binding technical architecture from which engineering, AI, infrastructure, and data specifications are derived. It is vendor-neutral and implementation-directive at the contract level.

CONTROL	VALUE
Product	The Eye
Tagline	The Operating System for Strategic Intelligence
Controlled document	Volume 3 — Technical Architecture
Edition	1.0
Issue date	1 August 2026
Status	Controlled Technical Architecture Baseline
Binding sources	Volume 0 — Product Constitution v1.0; Volume 1 — Executive Vision Book v1.0
Primary audience	Founders, CTO, enterprise and solution architects, principal engineers, AI and data leaders, security, governance, SRE, product and programme leadership
Normative scope	Logical layers, components, canonical objects, interfaces, runtime flows, trust zones, control planes, deployment invariants, quality attributes, and architecture governance
Downstream documents	Volume 4 — Engineering Specification; Volume 5 — AI Architecture; Volume 6 — Infrastructure Architecture; Volume 7 — Data Platform

Normative language

Shall and must indicate mandatory conformance. Should indicates the expected enterprise implementation unless an approved exception demonstrates equivalent behavior. May indicates an allowed implementation choice behind a frozen contract. Architecture diagrams are normative for boundaries and relationships; named technologies are intentionally absent.

> **CONTROLLING RULE**
> Where this volume conflicts with the Product Constitution, Volume 0 controls. Where a downstream specification conflicts with this volume, Volume 3 controls until an approved architecture or constitutional change is issued.

CONSTITUTIONAL INHERITANCE / C-006 · C-008 · C-051 · C-052

Architecture Charter

The Eye is designed as a long-lived enterprise operating system for strategic intelligence, not as a collection of AI features. Architecture therefore protects meaning, authority, memory, evidence, and institutional continuity before optimizing any single implementation.

Frozen architecture commitments

COMMITMENT	ARCHITECTURE CONSEQUENCE
One closed strategic loop	Observe, Understand, Predict, Simulate, Decide, and Learn operate through canonical objects and governed re-entry, not disconnected applications.
Ten canonical layers	Each layer owns a distinct responsibility, state boundary, interface surface, control obligations, and degradation behavior.
Knowledge Graph centrality	Shared identity, semantics, relationships, time, provenance, and Strategy Graph context remain connected across the platform.
Human decision sovereignty	AI may recommend, forecast, simulate, reason, explain, and prepare; a named human retains final authority for consequential decisions.
Explainability by construction	Evidence, assumptions, uncertainty, alternatives, versions, policy, and accountable action are captured throughout production, not generated after the fact.
Deployment semantic parity	SaaS, Private Cloud, and On-Premise preserve one object model, workflow, trust model, authority model, and portability contract.
Continuous governed learning	Outcomes create proposals evaluated and released through governance; production behavior does not silently self-modify.
Customer control	Data, keys, models, policy, administration, residency, export, deletion, and exit follow declared customer and legal boundaries.

Architecture qualities

- Semantically coherent across layers, domains, customers, and deployment modes.
- Temporally reconstructable without hindsight contamination.
- Provenance-complete from source evidence through decisions and outcomes.
- Secure, private, sovereign, resilient, observable, portable, and governable by design.
- Model- and vendor-plural while preserving one strategic product contract.
- Contestable by humans at every material analytical and decision boundary.

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Contents

The document uses native Word heading styles for navigation. Chapter numbering is stable and referenced by downstream volumes.

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Executive Architecture Statement

The Eye is a strategic state, reasoning, experimentation, decision, and learning system organized around one governed representation of the world and the enterprise.

The architecture begins at the evidence boundary and ends only after outcomes return to institutional memory. It preserves what was observed, what was claimed, what was inferred, what humans assessed, which futures were considered, what simulations assumed, which recommendation was made, who decided, what the institution committed to do, what occurred, and what learning was later approved.

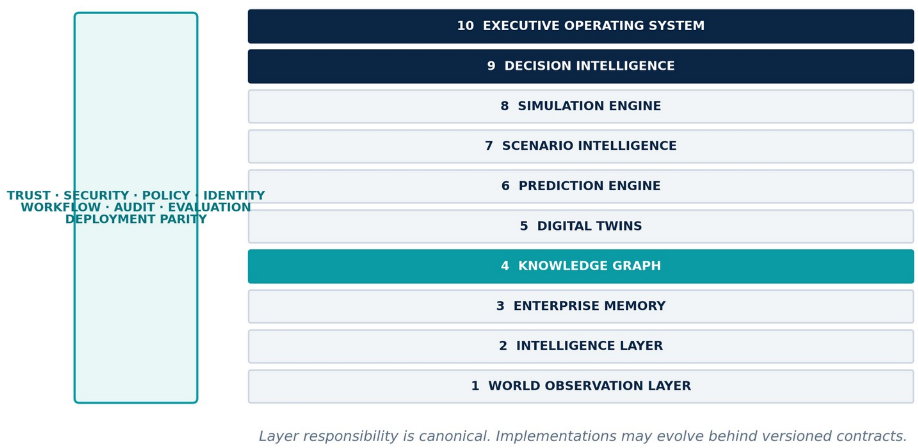

Figure 1. The ten canonical layers with shared cross-cutting control planes.

The operating structure

ARCHITECTURE PLANE	PURPOSE
Ten canonical layers	Own the strategic lifecycle from observation through executive operation.
Knowledge and state substrates	Enterprise Memory, Knowledge Graph, Strategy Graph, Digital Twins, temporal truth, and canonical objects preserve connected institutional state.
Foresight and decision system	Prediction, scenarios, simulation, decision packages, commitments, and Decision Replay make uncertainty actionable without transferring final authority.
Control planes	Trust, provenance, security, privacy, identity, policy, workflow, audit, evaluation, learning, deployment, and sovereignty apply across every layer.
Execution fabric	Commands, queries, events, streams, workflows, artifact packages, agents, models, and external gateways coordinate reliable behavior.
Deployment envelope	SaaS, Private Cloud, and On-Premise bind the same semantics to different physical, operational, and sovereignty boundaries.

> **ARCHITECTURE THESIS**
> The Eye does not make strategy autonomous. It makes strategic work persistent, connected, testable, explainable, accountable, and capable of learning across decisions and generations of leadership.

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PART I

System Definition

Authority, context, lifecycle, canonical layers, control planes, boundaries, objects, and interaction contracts.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

1 Authority, Scope, and Architecture Status

Volume 3 is the authoritative logical and operational architecture for The Eye. It converts the Product Constitution and Executive Vision into stable system boundaries, component responsibilities, contracts, flows, controls, and measurable architecture behavior.

Normative authority

This volume inherits Volume 0 — Product Constitution v1.0 and Volume 1 — Executive Vision Book v1.0. Where explanatory language could be interpreted in more than one way, the constitutional invariant is controlling. Architecture may elaborate an invariant; it may not weaken, rename, collapse, or silently reinterpret it.

The architecture is frozen at the level defined here: the ten layers, their canonical responsibilities, the shared object and control semantics, the closed strategic loop, Knowledge Graph centrality, human decision sovereignty, and semantic parity across SaaS, Private Cloud, and On-Premise. Technology products, capacity figures, deployment topology details, and implementation mechanisms are specified in later volumes without changing these contracts.

- Shall and must indicate mandatory conformance.
- Should indicates the normal enterprise implementation unless an approved exception demonstrates equivalent behavior.
- May indicates a permitted implementation choice behind the frozen contract.
- Cannot and must not identify prohibited behavior or a constitutional boundary.

Architecture scope

IN SCOPE	CONTROLLED BY THIS VOLUME
System structure	Ten layers, cross-cutting planes, logical components, trust zones, and customer intelligence domains.
Information structure	Canonical objects, truth states, temporal semantics, provenance, identity, policy, and versioning.
Behavior	Commands, queries, events, workflows, human gates, corrections, replay, degradation, and recovery.
Deployment invariants	Semantic parity, tenancy, sovereignty boundaries, portability, and disconnected operating envelopes.
Governance	Decision rights, ADRs, conformance, exceptions, quality attributes, evolution, and verification.

Deliberate boundaries

Volume 3 does not select vendors, programming languages, databases, orchestration products, foundation models, cloud providers, or hardware. It defines what an implementation must preserve when those choices are made. Volume 4 specifies engineering behavior and interfaces; Volume 5 specifies AI and agent architecture; Volume 6 specifies infrastructure; Volume 7 specifies the data platform.

> **ARCHITECTURE RULE**
> An implementation detail is replaceable only when identity, semantics, time, provenance, policy, quality, auditability, and externally observable behavior remain intact.

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2 System Context and Stakeholders

The Eye sits between the world, the enterprise, strategic models, and accountable human authority. It receives evidence and purpose, produces governed intelligence and decision packages, and records what institutions chose and learned.

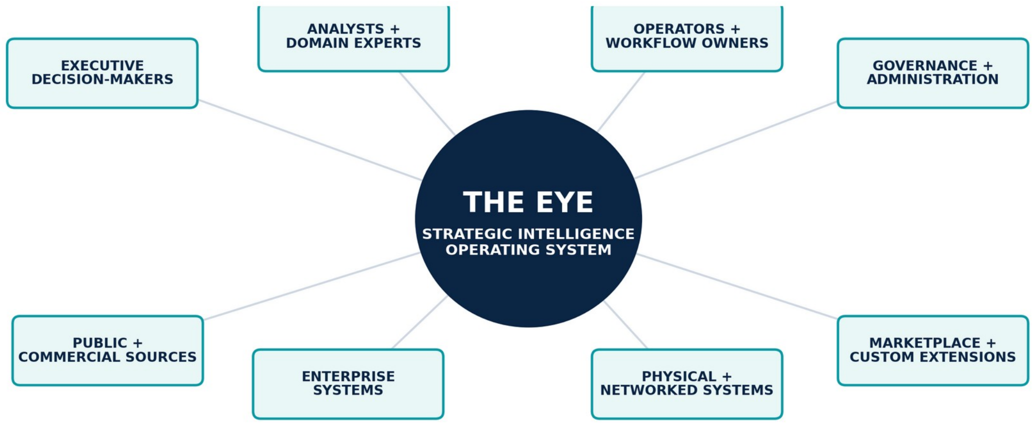

Figure 2. System context showing human authorities, enterprise and external systems around The Eye.

Primary human actors

ACTOR	ARCHITECTURE RESPONSIBILITY	AUTHORITY BOUNDARY
Executive decision-maker	Sets purpose, objectives, risk appetite, review cadence, and final commitment.	Final authority for consequential strategic choices.
Strategic analyst	Investigates evidence, constructs assessments, forecasts, scenarios, simulations, and decision packages.	May prepare and recommend; cannot silently approve.
Domain expert	Validates domain semantics, assumptions, models, constraints, mechanisms, and explanations.	Approves within delegated expertise; authority remains explicit.
Operator / workflow owner	Executes approved commitments, reports operational state, and owns response playbooks.	May act within approved workflow and policy envelope.
Governance, risk, legal, and compliance	Defines policy, review, retention, privacy, classification, and exception obligations.	May require, deny, constrain, review, or revoke governed actions.
Platform administrator and SRE	Operates identity, tenancy, deployment, reliability, recovery, and observability controls.	Cannot reinterpret strategic content or bypass decision authority.
Data / ontology steward	Owns source, quality, identity, ontology, graph, retention, and correction stewardship.	May govern information integrity without deciding strategy.

External systems and environments

- Public, commercial, scientific, financial, geopolitical, regulatory, climate, satellite, and social sources.
- ERP, CRM, finance, supply-chain, manufacturing, logistics, healthcare, telecom, and other enterprise systems.
- IoT, telemetry, geospatial, infrastructure, cyber, and physical operational systems.
- Identity providers, key-management systems, policy authorities, audit systems, model providers, and notification channels.
- Customer-defined connectors, domain packs, Agent Marketplace packages, and Scenario Marketplace packages.

Context boundary

The Eye may read from, write to, or trigger external systems only through declared contracts, authenticated identities, least-privilege permissions, policy evaluation, provenance, and audit. External execution is never inferred from a recommendation. A recommendation becomes an executable commitment only through an approved workflow owned by a named human authority.

> **HUMAN SOVEREIGNTY**
> Machine reach may extend observation, analysis, forecasting, simulation, and preparation. Purpose, policy, risk appetite, approval, final commitment, and high-impact rights remain human responsibilities.

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3 The Strategic Intelligence Lifecycle

Observe, Understand, Predict, Simulate, Decide, and Learn are not product modules arranged in a sequence. They are one governed lifecycle in which every artifact remains connected to its evidence, assumptions, methods, authority, decision state, and outcome.

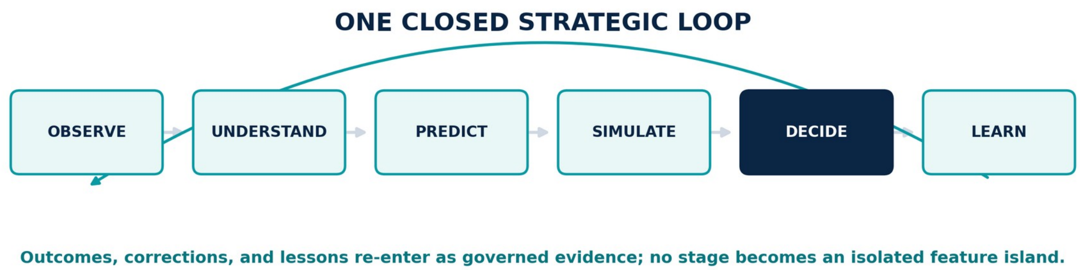

Figure 3. Closed strategic intelligence lifecycle from Observe through Learn, with outcomes re-entering as evidence.

Lifecycle contracts

STAGE	ENTRY CONDITION	OWNED TRANSFORMATION	EXIT CONDITION
Observe	Authorized source and purpose	Acquire and preserve governed evidence	Evidence has identity, time, rights, quality, and custody
Understand	Evidence is available and policy-cleared	Extract, resolve, relate, assess, and retain truth states	Structured understanding is reviewable and attributable
Predict	Target, horizon, context, and model set are declared	Estimate distributions, drivers, calibration, and sensitivity	Forecast Package is versioned, evaluated, and expiring
Simulate	Twin, scenario, intervention, constraints, and versions resolve	Run reproducible experiments and sensitivity	Results carry validation and challenge evidence
Decide	Objectives, options, evidence, uncertainty, and authority are complete	Compare, explain, route approval, and commit	Decision record and monitoring contract are explicit
Learn	Outcome, feedback, override, or correction is observed	Diagnose and propose governed changes	Approved release is monitored and reversible

Re-entry and nonlinearity

The lifecycle is logically closed but operationally nonlinear. A simulation may request new observations. A contradiction may reopen entity resolution. A forecast-fitness failure may invalidate scenario branches. A policy change may reopen a decision. An outcome may generate a lesson proposal rather than an immediate model update.

Every re-entry carries correlation and causation identifiers, object versions, policy context, and the state transition that caused the return. This makes the loop explainable and prevents invisible recursive behavior.

- No stage may erase the evidence or uncertainty of an earlier stage.
- No later outcome may rewrite what was knowable at decision time.
- No learning proposal may bypass evaluation, approval, release, monitoring, and rollback.
- No access mode may create a parallel lifecycle or separate truth.

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4 The Canonical Ten-Layer Architecture

The ten named layers are the stable product structure. Each layer owns a canonical responsibility and publishes governed interfaces. Technology and internal topology may change; the layer contract does not.

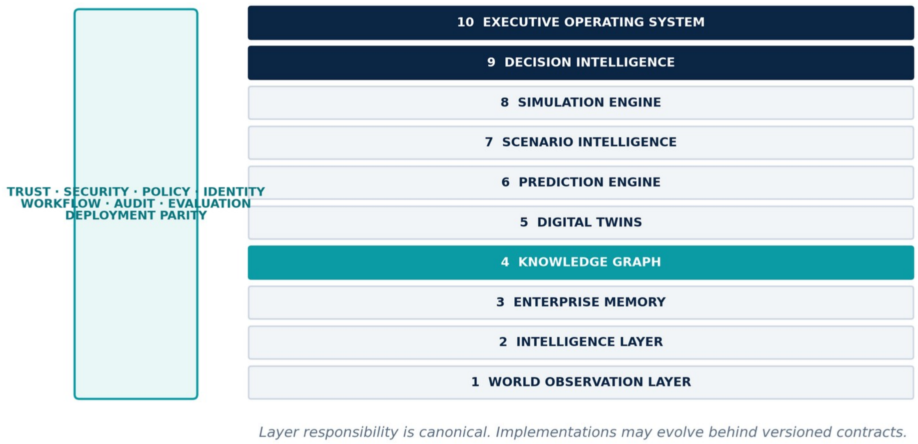

Figure 4. The ten canonical layers with shared cross-cutting control planes.

Layer contract model

A layer is a responsibility boundary, not a deployable unit. One layer may contain several services and stores; one service may implement supporting behavior for more than one layer only when ownership, interfaces, policy, data, and telemetry remain explicit. Physical deployment may co-locate or separate components without collapsing the logical architecture.

CONTRACT ELEMENT	MANDATORY MEANING
Responsibility	The outcome the layer alone is accountable for producing.
Inputs and outputs	Canonical objects or control decisions crossing the boundary.
Owned state	Authoritative records the layer versions, governs, and recovers.
Interfaces	Versioned commands, queries, events, artifact packages, and workflow transitions.
Controls	Identity, policy, provenance, quality, audit, safety, privacy, and human gates.
Failure behavior	How the layer abstains, degrades, contains, recovers, and signals impact.
Quality contract	Measurable product behavior for correctness, latency, freshness, reliability, cost, and trust.

Inter-layer discipline

- Layers exchange canonical objects; they do not infer meaning from storage tables or private model state.
- Commands request owned behavior; domain events announce committed state changes; queries return policy-filtered projections.
- Every material interface carries the common contract envelope defined in Chapter 8.
- A layer may reject, abstain, or route for review when its entry conditions are not met.
- Corrections propagate through subscriptions and impact analysis rather than destructive cross-layer writes.

REPLACEABILITY RULE

Internal implementations are replaceable. Canonical object identity, temporal and truth semantics, provenance, policy, audit, and observable behavior are not.

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5 Cross-Cutting Control Planes

Trust, security, policy, identity, workflow, evaluation, audit, sovereignty, and operations are continuous system behavior. None is delegated to a downstream compliance layer.

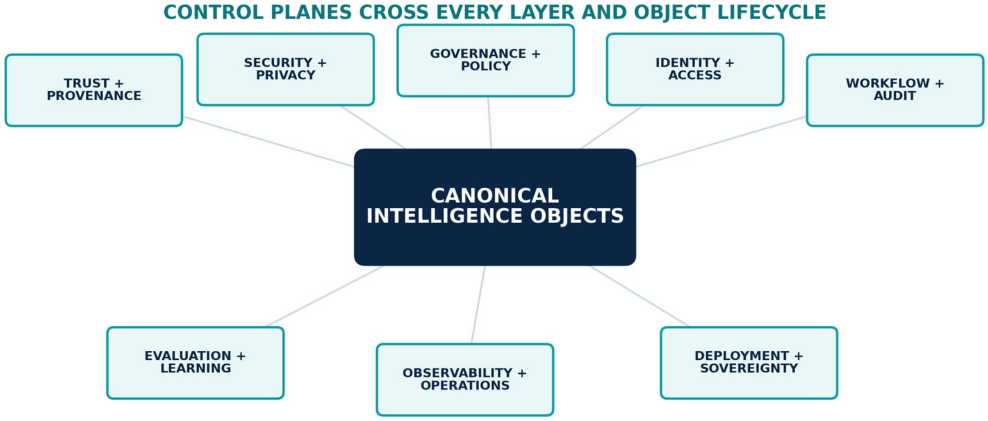

Figure 5. Cross-cutting control planes surrounding canonical intelligence objects.

Control-plane responsibilities

PLANE	RESPONSIBILITY	EVIDENCE PRODUCED
Trust and Provenance	Establishes source identity, chain of custody, transformation lineage, object-level evidence links, quality, and explanation paths.	Source, custody, transformation, object lineage, explanation, and correction records.
Security and Privacy	Enforces zero-trust access, classification, encryption, purpose limitation, minimization, threat controls, and safe handling.	Authentication, authorization, data handling, threat, consent, minimization, and protection evidence.
Governance and Policy	Turns constitutional, legal, customer, and institutional rules into versioned, executable controls and obligations.	Policy versions, decisions, obligations, denials, exceptions, expiry, and revocation.
Identity and Access	Authenticates human, device, workload, model, agent, source, and tenant identities; authorizes least-privilege actions continuously.	Identity binding, token, role, relationship, purpose, device, workload, and session decisions.
Workflow and Audit	Coordinates stateful work, human gates, separation of duties, retries, compensation, escalation, and non-repudiable execution evidence.	State transitions, checkpoints, retries, compensation, approvals, overrides, and actor attribution.
Evaluation and Learning	Measures source, extraction, model, forecast, scenario, simulation, agent, recommendation, and institutional performance; governs change proposals.	Metrics, test suites, fitness decisions, change proposals, releases, monitoring, and rollback.
Deployment and Sovereignty	Preserves semantic parity while applying tenancy, residency, key ownership, network, model, administrative, and disconnected-operation constraints.	Placement, residency, keys, network, model, administration, export, and disconnected-state evidence.

Inline enforcement

Control evaluation occurs before and after material actions. Before execution, the platform determines whether the identity, purpose, object, data, model, agent, workflow state, deployment boundary, and requested side effects are permitted. After execution, it records what occurred, checks postconditions, publishes telemetry, and initiates compensation or containment when behavior diverges.

Control decisions are themselves versioned objects. A denial, exception, override, or human approval is not a log message; it is governed evidence connected to the action and later decision replay.

NO TRUST BY LOCATION

Network location, deployment mode, model provider, agent role, or internal service identity never creates implicit trust. Authorization is continuous, contextual, least-privilege, and attributable.

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6 System Boundaries and Trust Zones

The architecture isolates untrusted evidence, customer strategic context, control functions, marketplace code, models, administrative operations, and exports. Every crossing is authenticated, authorized, policy-evaluated, provenance-bearing, and auditable.

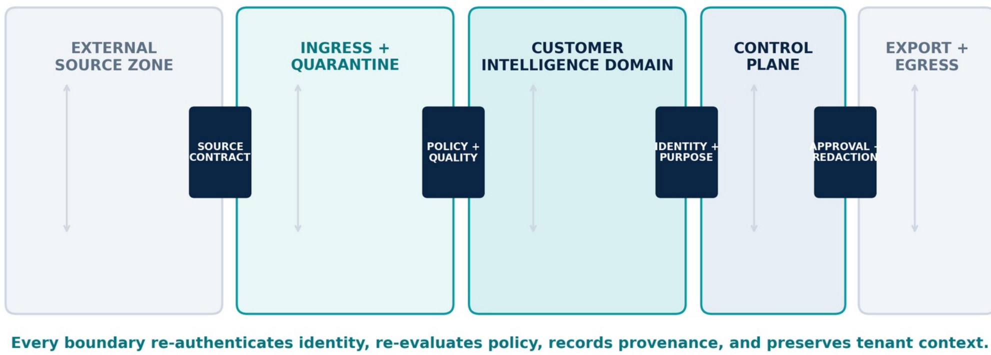

Figure 6. Trust-zone architecture from external sources through customer intelligence domains to controlled egress.

Customer Intelligence Domain

The Customer Intelligence Domain is the primary isolation boundary for a customer's evidence, graph, memory, twins, models, agents, policies, workflows, decisions, and audit records. It is a logical boundary in all deployment modes and may be implemented as one or more isolated cells, accounts, projects, namespaces, clusters, networks, or physical environments.

The domain carries a stable tenant identity through every object and action. Cross-domain operations are denied by default and require a declared federation, legal basis, policy, owner, purpose, object scope, transformation, and revocation path.

Boundary classes

BOUNDARY	DEFAULT POSTURE	REQUIRED GATE
External source → ingress	Untrusted	Source contract, authenticity, rights, malware, schema, classification, and quarantine.
Ingress → customer domain	Denied until qualified	Tenant binding, purpose, quality, provenance, policy, and normalization.
Service → service	No implicit trust	Workload identity, action scope, object policy, correlation, and audit.
Model / agent → tools and data	Capability denied by default	Purpose, owner, tool, data scope, budget, side-effect, and evaluation policy.
Customer domain → control plane	Metadata-minimized	Operational necessity, classification, minimization, regional and contractual policy.
Customer domain → export / execution	Human and policy gated	Approval, redaction, format, destination, data rights, audit, and revocation where supported.

Marketplace and sandbox boundary

Agent and Scenario Marketplace packages execute in a sandbox with explicit declared dependencies and no ambient access to customer memory, models, keys, networks, or external side effects. Capability grants are instance-specific, revocable, monitored, and constrained by the customer's deployment and policy boundary.

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7 Canonical Intelligence Object Model

The Eye is an object system before it is an interface. Every layer reads, writes, versions, relates, evaluates, and governs canonical intelligence objects with stable identity, temporal truth, provenance, policy, and lifecycle state.

THE CANONICAL LINEAGE NEVER BREAKS AT MODEL, AGENT, OR HUMAN BOUNDARIES

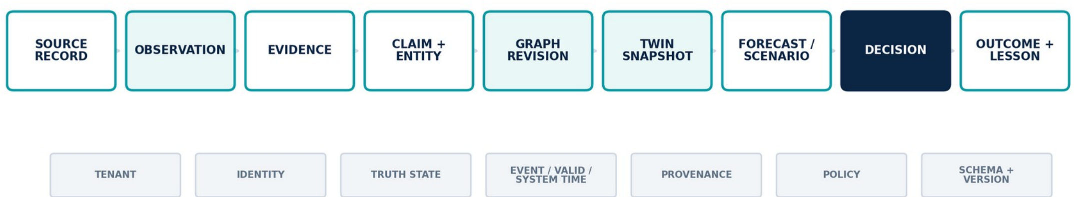

Figure 7. Canonical object lineage from source record to outcome and governed lesson.

Universal object header

FIELD GROUP	REQUIRED FIELDS	PURPOSE
Identity	object_id, object_type, tenant_id, domain_id	Stable identity and isolation.
Version	schema_version, object_version, supersedes, lifecycle_state	Evolution, correction, replay, and compatibility.
Time	event_time, observation_time, valid_from/to, system_from/to	Historical reconstruction without hindsight contamination.
Truth	truth_state, confidence, uncertainty, synthetic_state	Separates observation, claim, inference, human judgment, and scenario data.
Provenance	source_refs, evidence_refs, transformation_refs, model_refs, agent_refs	Backward explanation and forward impact analysis.
Control	classification, purpose, policy_refs, owner, approvals	Authorized and accountable use.
Quality	freshness, completeness, fitness, expiry, evaluation_refs	Measurable product behavior and safe degradation.
Operations	correlation_id, causation_id, idempotency_key, trace_id	Reliable flow, replay, compensation, and observability.

Canonical object families

Objects are immutable by version. A correction, withdrawal, adjudication, approval, or outcome creates a new version or linked object and preserves the prior record. Projections may present a current view, but the authoritative history remains reconstructable.

CODE	OBJECT	OWNING LAYER	DEFINITION
SRC	Source Contract	Observation	Authority, rights, purpose, schema, latency, quality, authenticity, correction behavior, and ownership for a source.
OBS	Observation Record	Observation	An immutable record that a source produced content or state at a specific observation time.
EVD	Evidence Object	Observation / Intelligence	Preserved source material plus chain of custody, transformations, classification, quality, and coverage context.

CODE	OBJECT	OWNING LAYER	DEFINITION
CLM	Claim	Intelligence	A source-attributed assertion with truth state, confidence, temporal scope, and supporting or contradicting evidence.
EVT	Event	Intelligence	A temporally scoped occurrence connected to actors, locations, assets, mechanisms, and evidence.
ENT	Entity	Knowledge Graph	A governed identity representing a company, country, person, product, technology, market, asset, objective, or other canonical thing.
REL	Relationship	Knowledge Graph	A typed, temporal, provenance-bearing connection between governed entities or intelligence objects.
ASM	Assessment	Intelligence	A versioned analytical judgment that separates evidence, method, uncertainty, dissent, and accountable author.
MEM	Memory Entry	Enterprise Memory	A durable, permission-aware, versioned organizational record with retention and correction semantics.
GRV	Graph Revision	Knowledge Graph	An atomic, validated set of node, edge, ontology, provenance, and temporal changes.
OBJ	Objective	Strategy Graph	A governed strategic intent with owner, horizon, measures, constraints, dependencies, and review cadence.
RSK	Risk / Opportunity	Strategy Graph	A symmetric strategic exposure connected to drivers, affected objectives, scenarios, responses, and owners.

Promotion rules

- Evidence is preserved; derived claims link to it rather than replacing it.
- Claims become accepted graph assertions only through ontology, identity, confidence, policy, and review rules.
- Model or agent output remains inferred or synthetic until an authorized process changes its truth state.
- A recommendation remains advisory until a named human authority creates a decision record and commitment.
- A lesson remains proposed until evaluation, approval, release, monitoring, and rollback controls are satisfied.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-011 · C-012 · C-016 · C-025 · C-035

8 Interaction Styles and the Contract Envelope

The Eye combines synchronous queries, authoritative commands, domain events, evidence streams, workflow transitions, and portable artifact packages. Every interaction carries a common envelope so trust and meaning do not disappear at service boundaries.

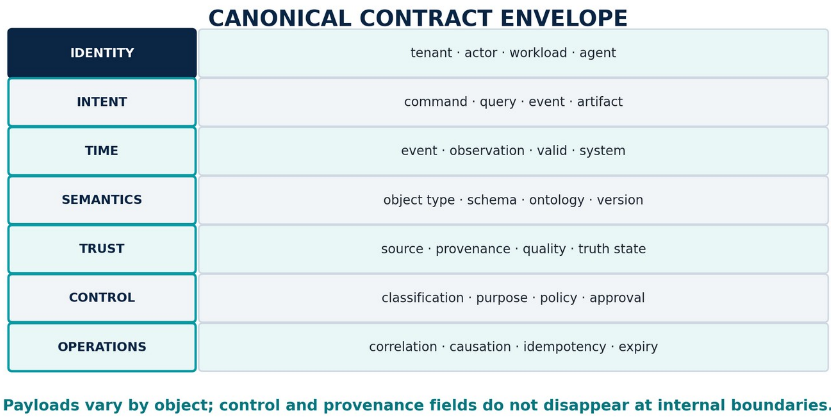

Figure 8. Canonical interface envelope carrying identity, intent, time, semantics, trust, control, and operational metadata.

Interaction semantics

STYLE	USE	OWNERSHIP AND RELIABILITY RULE
Command	Requests a state change from the component that owns the behavior.	Single accountable receiver; idempotent where retry is possible; produces success, rejection, or domain events.
Query	Retrieves a policy-filtered projection or explanation without changing canonical state.	No hidden write; result declares source revision, freshness, coverage, and partial-state markers.
Domain event	Announces an authoritative state transition that already committed.	Immutable event ID; ordered within aggregate or partition; consumers are idempotent and independently recoverable.
Evidence stream	Carries high-volume observations, telemetry, market, IoT, or change data.	Checkpointed, backpressured, replayable, classified, and tied to source contract and observation time.
Workflow transition	Advances long-running work with approvals, retries, waits, compensation, and human tasks.	Durable state machine; every transition has actor, policy, input versions, and audit evidence.
Artifact package	Moves signed scenarios, agents, models, twin configurations, briefings, or export bundles.	Manifested, versioned, inspectable, permission-scoped, validated, and revocable.

Compatibility and evolution

- Schemas use explicit semantic versions and machine-readable compatibility metadata.
- Additive evolution is preferred; breaking changes require migration, dual-read or dual-write strategy, rollback, and consumer verification.
- Events are facts and are never mutated; correction is represented by a new event linked through causation and supersession.
- Queries declare revision and consistency requirements instead of assuming a single global transaction.
- Contract tests, conformance fixtures, and consumer-driven verification are release gates.

CONTRACT RULE
A service boundary is not governed if the payload is versioned but identity, time, provenance, truth state, policy context, and causation are missing.

CONSTITUTIONAL INHERITANCE / C-006 · C-011 · C-012 · C-035 · C-038 · C-043 · C-049

PART II

Canonical Layer Architecture

The ten frozen layers, their responsibilities, owned components, interfaces, controls, and failure behavior.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

9 World Observation Layer

Maintain a measurable, governed field of view across authorized external, internal, physical, and networked sources.

Responsibility contract

CONTRACT	DEFINITION
Owns	Maintain a measurable, governed field of view across authorized external, internal, physical, and networked sources.
Boundary	Owns source contracts, acquisition, raw preservation, source health, coverage, and correction intake. It does not infer institutional truth or publish recommendations.
Authoritative state	Source contracts and credential references; Connector configuration and checkpoints; Immutable raw evidence; Acquisition and source-health history; Coverage map and blind spots.
Quality dimensions	coverage, freshness, latency, completeness, authenticity, collection success, correction propagation time.

Inputs and outputs

INPUTS	OUTPUTS
• Authorized source definitions • Credentials and collection authority • Schedules, subscriptions, and stream endpoints • Customer purpose and coverage requirements	• Observation Records • Evidence Objects • Source health and freshness state • Coverage gaps • Correction and withdrawal events

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L1-C01	Source Registry and Contract Service	Owns versioned source registry and contract definitions, resolution, compatibility, stewardship state, and lifecycle events within World Observation Layer.
L1-C02	Connector Runtime and Adapter SDK	Mediates connector runtime and adapter sdk through authenticated, versioned, policy-evaluated contracts without leaking external semantics into the layer core.
L1-C03	Collection Scheduler and Subscription Manager	Coordinates collection scheduler and subscription lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L1-C04	Streaming and Batch Receivers	Owns the streaming and batch receivers capability inside World Observation Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L1-C05	Raw Preservation and Evidence Vault	Persists authoritative raw preservation and evidence vault state with version, integrity, policy, retention, and recovery semantics.
L1-C06	Authenticity and Chain-of-Custody Service	Owns the authenticity and chain-of-custody service capability inside World Observation Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L1-C07	Quarantine and Malware Inspection	Owns the quarantine and malware inspection capability inside World Observation Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L1-C08	Source Health, Freshness, and Coverage Monitor	Measures source health, freshness, and coverage monitor health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.
L1-C09	Correction and Withdrawal Intake	Owns the correction and withdrawal intake capability inside World Observation Layer, including contract validation, versioned behavior, operational evidence, and failure state.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L1-I01	RegisterSource	Command	Creates or versions a governed source contract after authorization and rights checks.
L1-I02	Acquire	Command / stream	Collects content or state under contract, purpose, rate, and residency constraints.
L1-I03	ObservationRecorded	Domain event	Announces an immutable observation and evidence reference.
L1-I04	SourceHealthChanged	Domain event	Reports freshness, completeness, authenticity, latency, or coverage state changes.
L1-I05	CorrectionReceived	Domain event	Introduces source correction, withdrawal, or supersession without destroying prior history.

Control obligations

- Purpose and legal authority before collection
- Tenant, classification, and residency assignment at ingress
- Rate, robots, licence, and contractual limits
- Authenticity and malware checks
- Immutable evidence identifiers and custody events

Degraded behavior

- Stale source state remains visible
- Derived products receive source-health impact markers
- Unsafe content stays quarantined
- No silent substitution of an unavailable source
- Coverage gaps enter warning and decision packages

LAYER ACCEPTANCE

World Observation Layer conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-012 · C-013 · C-014 · C-035 · C-039 · C-043

10 Intelligence Layer

Transform heterogeneous evidence into structured, reviewable understanding while preserving uncertainty, method, and source attribution.

Responsibility contract

CONTRACT	DEFINITION
Owns	Transform heterogeneous evidence into structured, reviewable understanding while preserving uncertainty, method, and source attribution.
Boundary	Owns deterministic and model-assisted transformation, extraction, classification, linking candidates, contradiction detection, assessments, and human review. It does not overwrite evidence or declare unreviewed model output as fact.
Authoritative state	Transformation definitions and versions; Candidate intelligence objects; Confidence and review state; Method and model lineage; Evaluation and adjudication history.
Quality dimensions	precision, recall, entity-link accuracy, contradiction detection, calibration, review yield, latency, cost, abstention quality.

Inputs and outputs

INPUTS	OUTPUTS
• Evidence Objects • Schemas and ontologies • Approved transformation and model versions • Domain policies and review thresholds	• Claims and Events • Entity and relationship candidates • Assessments and summaries • Contradictions and uncertainty • Transformation lineage and evaluation results

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L2-C01	Format Detection, OCR, Transcription, and Parsing	Owns the format detection, ocr, transcription, and parsing capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C02	Cleaning, Normalization, and Language Services	Owns the cleaning, normalization, and language services capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C03	Classification and Routing	Owns the classification and routing capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C04	Summarization and Structured Extraction	Owns the summarization and structured extraction capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C05	Named-Entity Recognition and Candidate Linking	Owns the named-entity recognition and candidate linking capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C06	Relationship, Event, and Claim Extraction	Owns the relationship, event, and claim extraction capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C07	Contradiction and Corroboration Detection	Owns the contradiction and corroboration detection capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C08	Confidence and Truth-State Assessor	Executes the governed confidence and truth-state assessor method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L2-C09	Human Review Queue and Adjudication	Coordinates durable human review queue and adjudication transitions, human tasks, deadlines, separation of duties, challenge, and audit evidence.

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L2-C10	Transformation Registry and Evaluation Harness	Owns versioned transformation registry and evaluation harness definitions, resolution, compatibility, stewardship state, and lifecycle events within Intelligence Layer.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L2-I01	TransformEvidence	Command	Runs a declared transformation plan against immutable evidence.
L2-I02	IntelligenceObject Proposed	Domain event	Publishes a claim, event, entity, relationship, or assessment candidate with lineage.
L2-I03	ContradictionDetected	Domain event	Links incompatible assertions without collapsing them into a single truth state.
L2-I04	ReviewRequested	Workflow event	Routes ambiguous, high-impact, low-confidence, or policy-sensitive output to accountable humans.
L2-I05	TransformationEvaluated	Domain event	Records quality, safety, cost, latency, and fitness for the producing version.

Control obligations

- Truth-state labels on every output
- Evidence references remain mandatory
- Model and prompt versions are explicit
- Human review thresholds follow risk and impact
- Derived objects never replace source evidence

Degraded behavior

- Abstain when evidence or model fitness is insufficient
- Route to deterministic fallback or human review
- Expose reduced confidence and unavailable methods
- Quarantine prompt-injection or malicious content
- Prevent downstream promotion when required review is missing

LAYER ACCEPTANCE

Intelligence Layer conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-011 · C-012 · C-030 · C-045

11 Enterprise Memory

Preserve durable, permission-aware organizational and world memory across evidence, analyses, decisions, actions, outcomes, and lessons.

Responsibility contract

CONTRACT	DEFINITION
Owns	Preserve durable, permission-aware organizational and world memory across evidence, analyses, decisions, actions, outcomes, and lessons.
Boundary	Owns persistence, version history, retention, legal hold, correction, retrieval indexes, and memory access policy. It does not convert retrieval frequency or model output into accepted organizational truth.
Authoritative state	Memory objects and versions; Retention and legal-hold state; Retrieval indexes; Correction and tombstone history; Archive, export, and deletion evidence.
Quality dimensions	durability, retrieval relevance, permission correctness, temporal completeness, reconstruction success, retention compliance, deletion verification.

Inputs and outputs

INPUTS	OUTPUTS
• Evidence and intelligence objects • Documents and communications • Graph, twin, decision, and outcome artifacts • Retention, deletion, and legal-hold policy	• Versioned memory entries • Permission-aware retrieval context • Historical reconstruction • Retention and deletion evidence • Strategic record packages

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L3-C01	Memory Object Service	Owns the memory object service capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C02	Document and Artifact Repository	Persists authoritative document and artifact repository state with version, integrity, policy, retention, and recovery semantics.
L3-C03	Temporal Record Store	Persists authoritative temporal record store state with version, integrity, policy, retention, and recovery semantics.
L3-C04	Version and Correction Ledger	Owns the version and correction ledger capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C05	Retention, Legal Hold, and Deletion Service	Owns the retention, legal hold, and deletion service capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C06	Permission-Aware Semantic Index	Owns the permission-aware semantic index capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C07	Memory Retrieval and Context Assembly	Owns the memory retrieval and context assembly capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C08	Archive and Cold-Tier Manager	Coordinates archive and cold-tier lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L3-C09	Memory Integrity and Orphan Detection	Owns the memory integrity and orphan detection capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L3-I01	CommitMemory	Command	Persists a governed object version with policy, provenance, classification, and temporal metadata.
L3-I02	RetrieveContext	Query	Returns policy-filtered memory and explanation links for an explicit purpose.
L3-I03	MemoryCorrected	Domain event	Versions a correction and identifies affected downstream objects.
L3-I04	RetentionActionDue	Domain event	Initiates archive, review, legal hold, deletion, or customer export workflow.
L3-I05	DeletionVerified	Audit event	Proves scope, authorization, execution, and residual references for a deletion action.

Control obligations

- Tenant and object-level access
- Purpose-aware retrieval
- Versioned corrections rather than destructive overwrite
- Lawful retention and verifiable deletion
- Isolation of unverified synthetic and model-generated content

Degraded behavior

- Fallback to metadata-only retrieval when content tier is unavailable
- Prevent stale index results from appearing current
- Expose partial reconstruction
- Pause deletion when referential scope cannot be proven
- Recover indexes from canonical records, not vice versa

LAYER ACCEPTANCE

Enterprise Memory conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-011 · C-015 · C-016 · C-025 · C-037 · C-040 · C-043

12 Knowledge Graph

Provide shared identity, semantics, temporal relationships, and provenance connecting the world, the enterprise, intelligence objects, and strategic intent.

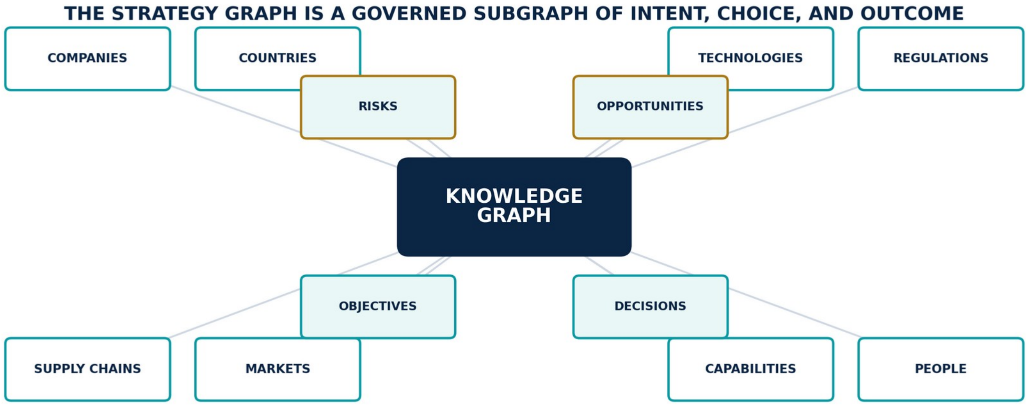

Figure 9. Knowledge Graph and governed Strategy Graph connecting world and strategic entities.

Responsibility contract

CONTRACT	DEFINITION
Owns	Provide shared identity, semantics, temporal relationships, and provenance connecting the world, the enterprise, intelligence objects, and strategic intent.
Boundary	Owns ontology governance, entity identity, relationship validation, temporal graph state, graph revisions, provenance edges, and graph subscriptions. It is the semantic heart, not the only physical store or computation engine.
Authoritative state	Ontology and semantic versions; Canonical entity identities; Temporal nodes and edges; Provenance links; Graph revision and stewardship history.
Quality dimensions	identity precision, identity recall, edge validity, provenance completeness, temporal completeness, orphan rate, query latency, revision propagation.

Inputs and outputs

INPUTS	OUTPUTS
• Entity and relationship candidates • Claims, events, and assessments • Ontology and policy changes • Corrections, merges, splits, and withdrawals	• Canonical entity identities • Temporal relationships • Graph revisions and subscriptions • Impact paths and dependency views • Strategy Graph context

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L4-C01	Ontology Registry and Governance Workflow	Owns versioned ontology registry and governance workflow definitions, resolution, compatibility, stewardship state, and lifecycle events within Knowledge Graph.
L4-C02	Identity Resolution and Entity Stewardship	Owns the identity resolution and entity stewardship capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.
L4-C03	Graph Write Validator	Owns the graph write validator capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L4-C04	Temporal Graph Store	Persists authoritative temporal graph store state with version, integrity, policy, retention, and recovery semantics.
L4-C05	Provenance and Evidence Edge Service	Owns the provenance and evidence edge service capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.
L4-C06	Graph Query and Traversal Service	Owns the graph query and traversal service capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.
L4-C07	Reasoning and Constraint Engine	Executes the governed reasoning and constraint method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L4-C08	Graph Change Subscription Service	Owns the graph change subscription service capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.
L4-C09	Merge, Split, and Reconciliation Manager	Coordinates merge, split, and reconciliation lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L4-C10	Graph Quality and Orphan Monitor	Measures graph quality and orphan monitor health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L4-I01	ResolveIdentity	Command / query	Returns or proposes canonical identity with evidence, confidence, and stewardship state.
L4-I02	CommitGraphRevision	Command	Applies an atomic, validated node, edge, ontology, and provenance change set.
L4-I03	GraphChanged	Domain event	Publishes affected identities, relationships, temporal scopes, and downstream subscriptions.
L4-I04	TraverseContext	Query	Returns policy-filtered paths with provenance, time, truth state, and confidence.
L4-I05	OntologyChange Proposed	Workflow event	Initiates compatibility, migration, domain, and governance review.

Control obligations

- No edge without type, time, provenance, and policy context
- Identity merges and splits are reversible
- Ontology changes require compatibility analysis
- Graph queries remain permission-aware
- Synthetic relationships are explicitly marked

Degraded behavior

- Serve last-known graph revision with staleness marker
- Queue writes when validation or consensus is unavailable
- Prevent partial multi-object revisions
- Expose unresolved identities rather than forcing links
- Rebuild projections from revision history

LAYER ACCEPTANCE

Knowledge Graph conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-007 · C-009 · C-010 · C-011 · C-012 · C-017 · C-035

13 Digital Twins

Maintain living, versioned strategic representations of the customer, competitors, supply chain, products, markets, regulations, infrastructure, processes, and organization.

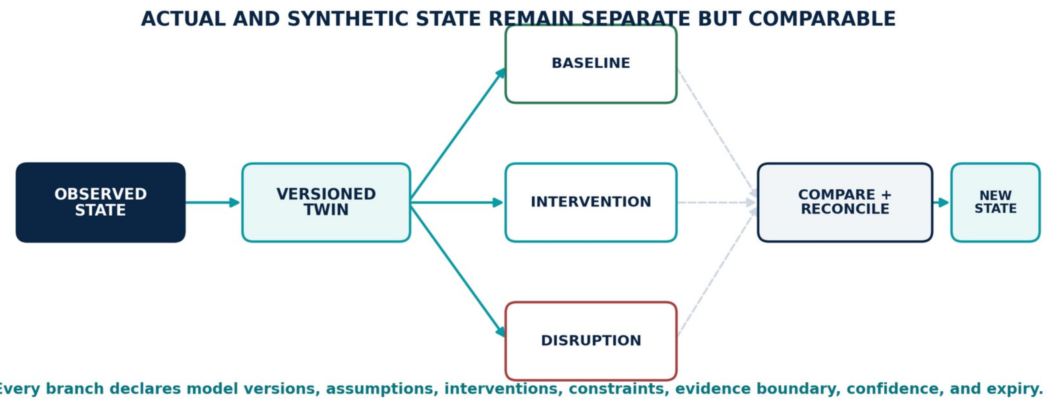

Figure 10. Digital Twin versioning, branching, comparison, and reconciliation model.

Responsibility contract

CONTRACT	DEFINITION
Owns	Maintain living, versioned strategic representations of the customer, competitors, supply chain, products, markets, regulations, infrastructure, processes, and organization.
Boundary	Owns twin definitions, evidence boundaries, state estimation, model bindings, snapshots, branches, reconciliation, validation, and dependency maps. It never presents simulated state as observed fact.
Authoritative state	Twin definitions and owners; Observed-state snapshots; Synthetic branches; Model and assumption bindings; Validation, confidence, and reconciliation history.
Quality dimensions	state freshness, estimation error, model fitness, coverage, reconciliation success, branch isolation, dependency completeness.

Inputs and outputs

INPUTS	OUTPUTS
• Graph identities and relationships • Observed and operational state • Approved behavioral and causal models • Assumptions, constraints, and validation rules	• Twin snapshots • Observed and synthetic branches • Dependency and behavior state • Validation and confidence state • Scenario and simulation inputs

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L5-C01	Twin Registry and Boundary Service	Owns versioned twin registry and boundary definitions, resolution, compatibility, stewardship state, and lifecycle events within Digital Twins.
L5-C02	State Estimation and Reconciliation	Owns the state estimation and reconciliation capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.
L5-C03	Model Binding and Operating-Envelope Registry	Owns versioned model binding and operating-envelope registry definitions, resolution, compatibility, stewardship state, and lifecycle events within Digital Twins.
L5-C04	Snapshot and Version Manager	Coordinates snapshot and version lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L5-C05	Branch and Time-Travel Manager	Coordinates branch and time-travel lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L5-C06	Twin Dependency Mapper	Owns the twin dependency mapper capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.
L5-C07	Behavior and Constraint Runtime	Owns the behavior and constraint runtime capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.
L5-C08	Validation and Calibration Service	Owns the validation and calibration service capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.
L5-C09	Synthetic-State Labelling and Isolation	Owns the synthetic-state labelling and isolation capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L5-I01	CreateTwinDefinition	Command	Declares boundary, identity, evidence, behaviors, owners, validation, and limitations.
L5-I02	ReconcileTwin	Command	Produces a new observed-state snapshot from evidence and graph revisions.
L5-I03	BranchTwin	Command	Creates an isolated synthetic branch for scenario or simulation use.
L5-I04	TwinStateChanged	Domain event	Publishes version, changed variables, confidence, freshness, and dependency impacts.
L5-I05	ValidateTwin	Command / event	Evaluates state and model fitness inside the declared operating envelope.

Control obligations

- Observed and synthetic state separation
- Exact evidence and graph revision per snapshot
- Versioned model and assumption bindings
- Declared operating envelope
- Simulation preserves the exact twin version used

Degraded behavior

- Freeze last validated snapshot with freshness warning
- Disable behaviors outside calibrated envelope
- Require reconciliation before merging a branch
- Expose missing variables and dependency uncertainty
- Prevent synthetic branch leakage into observed state

LAYER ACCEPTANCE

Digital Twins conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-010 · C-011 · C-018 · C-019 · C-023 · C-035 · C-043

14 Prediction Engine

Estimate future variables, events, and system states across the canonical 30-day, 90-day, 180-day, 1-year, 3-year, and 5-year horizons.

Responsibility contract

CONTRACT	DEFINITION
Owns	Estimate future variables, events, and system states across the canonical 30-day, 90-day, 180-day, 1-year, 3-year, and 5-year horizons.
Boundary	Owns forecast definitions, horizon methods, feature context, model routing, ensembles, calibration, uncertainty, sensitivity, evaluation, drift, and forecast versions. It does not claim certainty or select institutional action.
Authoritative state	Forecast definitions; Feature and model bindings; Forecast versions; Calibration and backtest history; Drift, expiry, and withdrawal state.
Quality dimensions	calibration, proper scoring rules, coverage, bias, sharpness, stability, drift, latency, cost, decision relevance.

Inputs and outputs

INPUTS	OUTPUTS
• Governed evidence and graph context • Twin snapshots • Forecast definitions and horizons • Approved models, features, and evaluation thresholds	• Forecast Packages • Probability distributions and confidence • Drivers and sensitivity • Calibration and fitness state • Refresh triggers and failure conditions

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L6-C01	Forecast Registry and Horizon Service	Owns versioned forecast registry and horizon definitions, resolution, compatibility, stewardship state, and lifecycle events within Prediction Engine.
L6-C02	Feature and Context Assembler	Owns the feature and context assembler capability inside Prediction Engine, including contract validation, versioned behavior, operational evidence, and failure state.
L6-C03	Model Gateway and Candidate Router	Mediates model gateway and candidate router through authenticated, versioned, policy-evaluated contracts without leaking external semantics into the layer core.
L6-C04	Ensemble and Disagreement Manager	Coordinates ensemble and disagreement lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L6-C05	Calibration and Uncertainty Service	Owns the calibration and uncertainty service capability inside Prediction Engine, including contract validation, versioned behavior, operational evidence, and failure state.
L6-C06	Driver Attribution and Sensitivity Engine	Executes the governed driver attribution and sensitivity method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L6-C07	Forecast Evaluation and Backtesting	Measures forecast evaluation and backtesting health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.
L6-C08	Drift and Refresh Monitor	Measures drift and refresh monitor health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.
L6-C09	Forecast Version Store	Persists authoritative forecast version store state with version, integrity, policy, retention, and recovery semantics.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L6-I01	RequestForecast	Command	Creates or refreshes a forecast under a declared objective, target, horizon, context, and policy.
L6-I02	ForecastProduced	Domain event	Publishes distribution, confidence, calibration, drivers, lineage, and expiry.
L6-I03	ForecastFitnessChanged	Domain event	Signals drift, calibration failure, data shift, or operating-envelope breach.
L6-I04	BacktestForecast	Command	Evaluates historical versions without contaminating prior decision context.
L6-I05	ForecastWithdrawn	Domain event	Marks a version unfit and propagates impact to scenarios, warnings, simulations, and decisions.

Control obligations

- Horizon-specific method selection
- Distribution rather than point estimate by default
- Model and feature lineage
- Calibration and failure conditions
- Alternative models and disagreement remain inspectable

Degraded behavior

- Abstain or widen uncertainty when fitness fails
- Serve prior forecast only with explicit expiry state
- Exclude unavailable model paths and disclose loss
- Prevent warnings from treating withdrawn forecasts as current
- Route material failures to scenario and decision owners

LAYER ACCEPTANCE
Prediction Engine conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-005 · C-020 · C-021 · C-030 · C-045 · C-049

15 Scenario Intelligence

Maintain a coherent, monitored portfolio of multiple plausible futures under uncertainty without allowing one forecast to masquerade as destiny.

Responsibility contract

CONTRACT	DEFINITION
Owns	Maintain a coherent, monitored portfolio of multiple plausible futures under uncertainty without allowing one forecast to masquerade as destiny.
Boundary	Owns scenario definitions, assumptions, actors, drivers, branches, indicators, signposts, interventions, coherence, comparison, review cadence, and portfolio state. It does not execute simulations or approve decisions.
Authoritative state	Scenario definitions and branches; Assumptions, actors, drivers, and mechanisms; Indicators and signposts; Coherence and confidence state; Review and retirement history.
Quality dimensions	assumption coverage, internal coherence, branch diversity, indicator freshness, signpost discrimination, review timeliness, decision coverage.

Inputs and outputs

INPUTS	OUTPUTS
• Forecast Packages • Twin snapshots and graph context • Objectives, assumptions, risks, and opportunities • Human and agent hypotheses	• Scenario branches and portfolios • Assumption registers • Indicators and signposts • Branch confidence and coherence state • Simulation and decision context

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L7-C01	Scenario Registry and Portfolio Manager	Owns versioned scenario registry and portfolio definitions, resolution, compatibility, stewardship state, and lifecycle events within Scenario Intelligence.
L7-C02	Assumption and Hypothesis Register	Owns versioned assumption and hypothesis register definitions, resolution, compatibility, stewardship state, and lifecycle events within Scenario Intelligence.
L7-C03	Branch and Path Engine	Executes the governed branch and path method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L7-C04	Actor, Driver, and Mechanism Model	Owns the actor, driver, and mechanism model capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L7-C05	Indicator and Signpost Service	Owns the indicator and signpost service capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L7-C06	Scenario Coherence Validator	Owns the scenario coherence validator capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L7-C07	Intervention and Impact Mapper	Owns the intervention and impact mapper capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L7-C08	Scenario Comparison and Review Workspace	Coordinates durable scenario comparison and review workspace transitions, human tasks, deadlines, separation of duties, challenge, and audit evidence.
L7-C09	Scenario Package Import and Localization	Owns the scenario package import and localization capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L7-I01	CreateScenario	Command	Declares baseline, branch logic, assumptions, drivers, owners, indicators, and review policy.
L7-I02	BranchScenario	Command	Creates a versioned upside, downside, disruption, or user-defined alternative.
L7-I03	ScenarioIndicator Changed	Domain event	Updates branch relevance when a monitored signpost moves.
L7-I04	ScenarioCoherence Failed	Domain event	Identifies internal inconsistency, invalid assumption, or incompatible dependency.
L7-I05	ScenarioReviewed	Workflow event	Records human review, dissent, continuation, retirement, or promotion to simulation.

Control obligations

- Multiple branches remain visible
- Assumptions are explicit and versioned
- Evidence and forecast dependencies are linked
- Indicators have owners and thresholds
- Imported packages cannot access customer memory by default

Degraded behavior

- Suspend a branch when critical assumptions become invalid
- Expose missing indicators and stale dependencies
- Prevent incoherent branches from entering simulation
- Retain retired branches for Decision Replay
- Route material branch movement to accountable owners

LAYER ACCEPTANCE

Scenario Intelligence conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-005 · C-022 · C-023 · C-032 · C-033 · C-046

16 Simulation Engine

Run reproducible experiments against declared twin versions, scenarios, assumptions, interventions, models, constraints, and seeds before institutional commitment.

Responsibility contract

CONTRACT	DEFINITION
Owns	Run reproducible experiments against declared twin versions, scenarios, assumptions, interventions, models, constraints, and seeds before institutional commitment.
Boundary	Owns experiment definitions, run orchestration, version resolution, constraints, solver adapters, impact analysis, sensitivity, validation, and reproducibility evidence. It does not silently alter production state or approve action.
Authoritative state	Experiment definitions; Resolved run manifests; Execution and resource state; Outputs and sensitivity; Validation, challenge, and invalidation history.
Quality dimensions	reproducibility, numerical stability, constraint satisfaction, sensitivity coverage, validation status, run latency, cost, failure containment.

Inputs and outputs

INPUTS	OUTPUTS
• Versioned twin and scenario branches • Interventions and constraints • Approved models and solver adapters • Run budgets, policies, and validation criteria	• Simulation Runs • Direct and second-order impacts • Sensitivity and uncertainty • Reproducibility package • Validation and challenge record

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L8-C01	Experiment Registry and Run Planner	Owns versioned experiment registry and run planner definitions, resolution, compatibility, stewardship state, and lifecycle events within Simulation Engine.
L8-C02	Twin, Scenario, and Model Version Resolver	Executes the governed twin, scenario, and model version resolver method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L8-C03	Constraint and Intervention Engine	Executes the governed constraint and intervention method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L8-C04	Seed and Determinism Manager	Coordinates seed and determinism lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L8-C05	Solver and Simulator Adapter Fabric	Mediates solver and simulator adapter fabric through authenticated, versioned, policy-evaluated contracts without leaking external semantics into the layer core.
L8-C06	Distributed Run Orchestrator	Coordinates distributed run orchestrator lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L8-C07	Impact and Second-Order Effect Analyzer	Executes the governed impact and second-order effect analyzer method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L8-C08	Sensitivity and Robustness Service	Owns the sensitivity and robustness service capability inside Simulation Engine, including contract validation, versioned behavior, operational evidence, and failure state.
L8-C09	Reproducibility and Validation Store	Persists authoritative reproducibility and validation store state with version, integrity, policy, retention, and recovery semantics.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L8-I01	SubmitExperiment	Command	Validates declared state, versions, constraints, budgets, policy, and expected outputs.
L8-I02	SimulationStarted	Domain event	Records resolved artifacts, execution identity, environment, seed, and run state.
L8-I03	SimulationCompleted	Domain event	Publishes outputs, impacts, uncertainty, sensitivity, validation, and resource evidence.
L8-I04	ChallengeSimulation	Workflow command	Re-runs or disputes assumptions, models, constraints, and interpretation.
L8-I05	SimulationInvalidated	Domain event	Marks results unfit and identifies dependent decision packages.

Control obligations

- Exact model, twin, scenario, and policy versions
- Seed and environment capture
- Resource budgets and stop conditions
- No write-back to observed state
- Reproducibility package accompanies material results

Degraded behavior

- Fail closed when required versions cannot resolve
- Return partial runs only with declared missing outputs
- Contain unstable solver adapters
- Preserve checkpoints and audit continuity
- Prevent invalid results from remaining decision-active

LAYER ACCEPTANCE

Simulation Engine conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-005 · C-018 · C-019 · C-023 · C-035 · C-043 · C-045

17 Decision Intelligence

Assemble complete, explainable choice packages and compare viable alternatives against objectives, constraints, obligations, risk, opportunity, uncertainty, reversibility, and information value.

PREDICTION INFORMS MULTIPLE FUTURES; SIMULATION TESTS INTERVENTION; HUMANS COMMIT

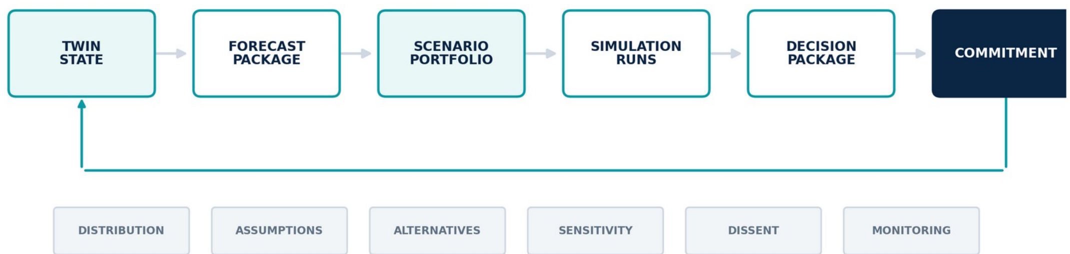

Figure 11. Foresight and decision flow from twin state through forecasts, scenarios, simulations, decisions, and commitments.

Responsibility contract

CONTRACT	DEFINITION
Owns	Assemble complete, explainable choice packages and compare viable alternatives against objectives, constraints, obligations, risk, opportunity, uncertainty, reversibility, and information value.
Boundary	Owns decision definitions, option context, evidence assembly, criteria, trade-offs, recommendation, dissent, approval routing, decision records, commitments, and monitoring contracts. A named human authority makes the final decision.
Authoritative state	Decision definitions and owners; Option and criteria versions; Decision package revisions; Approval, dissent, and override state; Final decisions, commitments, and monitoring contracts.
Quality dimensions	decision completeness, evidence coverage, alternative coverage, time-to-decision, reversibility, approval integrity, monitoring completeness, outcome linkage.

Inputs and outputs

INPUTS	OUTPUTS
• Objectives and constraints • Evidence, assessments, forecasts, scenarios, and simulations • Options and stakeholder obligations • Risk appetite, policy, and approval rules	• Decision Packages • Recommendations and alternatives • Approvals, dissent, overrides, and final decision record • Commitments and monitoring contracts • Decision Replay state

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L9-C01	Decision Registry and Context Resolver	Owns versioned decision registry and context resolver definitions, resolution, compatibility, stewardship state, and lifecycle events within Decision Intelligence.
L9-C02	Objective, Constraint, and Obligation Service	Owns the objective, constraint, and obligation service capability inside Decision Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L9-C03	Option and Alternative Manager	Coordinates option and alternative lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L9-C04	Evidence and Foresight Assembler	Owns the evidence and foresight assembler capability inside Decision Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L9-C05	Criteria, Trade-off, and Information-Value Engine	Executes the governed criteria, trade-off, and information-value method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L9-C06	Recommendation and Explanation Service	Owns the recommendation and explanation service capability inside Decision Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L9-C07	Approval, Dissent, and Override Workflow	Coordinates durable approval, dissent, and override workflow transitions, human tasks, deadlines, separation of duties, challenge, and audit evidence.
L9-C08	Decision Record and Commitment Service	Owns the decision record and commitment service capability inside Decision Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L9-C09	Monitoring Contract and Review Scheduler	Measures monitoring contract and review scheduler health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L9-I01	OpenDecision	Command	Declares question, accountable authority, scope, objectives, options, stakeholders, and required evidence.
L9-I02	DecisionPackageReady	Domain event	Publishes a complete reviewable package with uncertainty, alternatives, dissent, and provenance.
L9-I03	ApproveDecision	Human command	Records authenticated approval, rationale, obligations, overrides, and commitment state.
L9-I04	DecisionCommitted	Domain event	Activates commitments, execution handoff, monitoring conditions, and replay snapshot.
L9-I05	DecisionReopened	Workflow event	Re-enters the decision lifecycle when evidence, assumptions, policy, or monitoring conditions change.

Control obligations

- Named human authority
- Separation of preparation and approval
- Alternatives and dissent retained
- Recommendation criteria remain inspectable
- State of knowledge at decision time is immutable for replay

Degraded behavior

- Block commitment when mandatory evidence or approval is missing
- Expose stale or withdrawn inputs
- Allow human deferral, rejection, or request for information
- Prevent agent self-approval
- Reopen rather than silently rewriting a committed decision

LAYER ACCEPTANCE

Decision Intelligence conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-025 · C-026 · C-036 · C-050

18 Executive Operating System

Govern strategic attention, responsibility, review, commitments, scenarios, decisions, material change, and outcomes across institutional cadence.

Responsibility contract

CONTRACT	DEFINITION
Owns	Govern strategic attention, responsibility, review, commitments, scenarios, decisions, material change, and outcomes across institutional cadence.
Boundary	Owns executive workspaces, attention policy, briefing products, decision rooms, responsibility views, strategic cadence, notification, and access modes. It is not a passive dashboard and does not create separate truths for different roles.
Authoritative state	Attention and materiality policy; Briefing definitions and versions; Workspace and responsibility configuration; Review cadence and agenda state; Notification, suppression, delegation, and acknowledgement history.
Quality dimensions	time-to-understanding, decision latency, attention precision, warning acknowledgement, review completion, commitment visibility, accessibility, explanation use.

Inputs and outputs

INPUTS	OUTPUTS
• Material changes and warnings • Objectives, risks, opportunities, and scenarios • Pending decisions and commitments • Monitoring, outcomes, and review cadence	• Executive Briefings • Decision and review rooms • Prioritized attention queues • Commitment and outcome reviews • Strategic cadence and institutional state

Owned logical components

ID	LOGICAL COMPONENT	OWNED BEHAVIOR
L10-C01	Executive Workspace and Responsibility Model	Presents policy-filtered executive workspace and responsibility model products with evidence, uncertainty, authority, state, and required action intact.
L10-C02	Attention Policy and Materiality Engine	Executes the governed attention policy and materiality method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L10-C03	Live Briefing Service	Presents policy-filtered live briefing products with evidence, uncertainty, authority, state, and required action intact.
L10-C04	Decision and Scenario Rooms	Presents policy-filtered decision and scenario rooms products with evidence, uncertainty, authority, state, and required action intact.
L10-C05	Objective, Commitment, and Outcome Review	Coordinates durable objective, commitment, and outcome review transitions, human tasks, deadlines, separation of duties, challenge, and audit evidence.
L10-C06	Strategic Cadence Scheduler	Coordinates strategic cadence scheduler lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L10-C07	Notification and Escalation Service	Owns the notification and escalation service capability inside Executive Operating System, including contract validation, versioned behavior, operational evidence, and failure state.
L10-C08	Role-Appropriate Access Modes	Owns the role-appropriate access modes capability inside Executive Operating System, including contract validation, versioned behavior, operational evidence, and failure state.
L10-C09	Executive Search and Explanation	Owns the executive search and explanation capability inside Executive Operating System, including contract validation, versioned behavior, operational evidence, and failure state.

Interface surface

ID	INTERFACE	STYLE	CONTRACT
L10-I01	PublishBriefing	Command	Creates a live briefing bound to evidence, changes, objectives, decisions, and audience policy.
L10-I02	MaterialChangeRaised	Domain event	Routes change to accountable roles based on consequence, confidence, urgency, and attention policy.
L10-I03	ReviewConvened	Workflow event	Opens a governed review around a declared objective, decision, scenario, commitment, or outcome.
L10-I04	ExecutiveAction Requested	Human command	Requests analysis, scenario, simulation, decision, delegation, suppression, or follow-up.
L10-I05	AttentionPolicyChanged	Governance event	Versions materiality, escalation, suppression, and notification rules.

Control obligations

- Role and need-to-know access
- Materiality and notification policy
- Every displayed conclusion links to evidence and uncertainty
- Suppression and delegation remain visible
- Access mode never changes canonical meaning

Degraded behavior

- Expose incomplete briefings and unavailable dependencies
- Prefer silence over false certainty for unsafe products
- Retain urgent decision state during partial platform outage
- Preserve acknowledgement and escalation audit
- Allow controlled offline briefing packages where policy permits

LAYER ACCEPTANCE
Executive Operating System conforms only when its owned state, interface contracts, control obligations, quality dimensions, and degraded behavior remain independently testable in SaaS, Private Cloud, and On-Premise operation.

CONSTITUTIONAL INHERITANCE / C-003 · C-004 · C-027 · C-032 · C-033 · C-047 · C-048

PART III

Shared Intelligence Substrates

Strategy, temporal truth, provenance, policy, agents, models, workflows, evaluation, retrieval, marketplaces, and proactivity.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

19 Strategy Graph

The Strategy Graph connects institutional purpose to the world model. It makes objectives, assumptions, capabilities, risks, opportunities, initiatives, decisions, commitments, measures, and outcomes part of one governed strategic system.

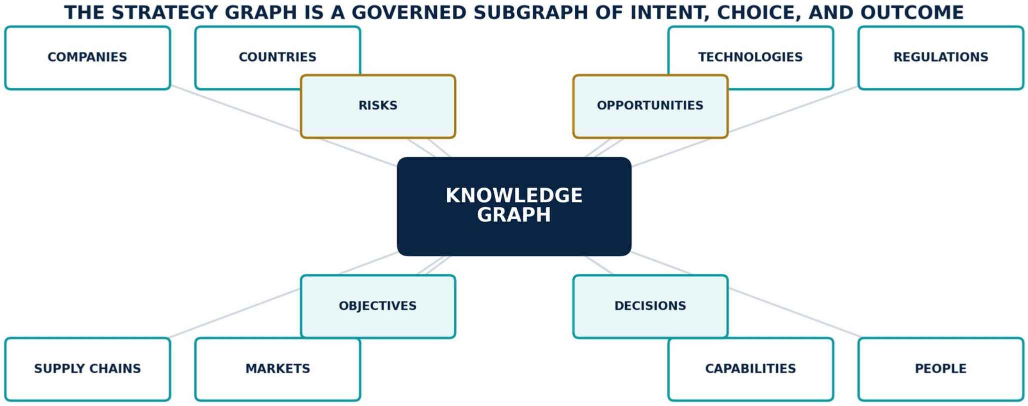

Figure 12. Knowledge Graph and governed Strategy Graph connecting world and strategic entities.

Strategic semantic model

The Strategy Graph is a governed subgraph of the Knowledge Graph, not a separate planning database. It uses the same canonical identities, temporal semantics, evidence links, policy controls, revisions, and explanation paths as the wider platform. Its distinguishing property is accountable strategic intent: every material object has an owner, horizon, status, review cadence, and relationship to objectives.

Objectives express intended outcomes. Assumptions express propositions that must remain true. Capabilities express what the institution can reliably do. Risks and opportunities express asymmetric exposure. Initiatives and commitments express approved action. Measures and outcomes show whether reality moved as intended. Decisions connect the state of knowledge to the commitments that followed.

Canonical relationship classes

RELATIONSHIP	MEANING	REQUIRED CONTROL
supports / conflicts with	Contribution or tension between objectives, capabilities, initiatives, and evidence.	Direction, strength, owner, evidence, time, and review.
depends on	A condition, capability, supplier, asset, policy, assumption, or prior commitment required for success.	Criticality, substitutability, lead time, and failure indicator.
exposes / mitigates	A risk or opportunity affects an objective; a response changes the exposure.	Mechanism, consequence, likelihood, response owner, and residual exposure.
decided by / commits	A decision authorizes, rejects, defers, or constrains an institutional commitment.	Human authority, rationale, approval, obligations, and replay snapshot.
measured by / evidenced by	A measure or outcome supports judgment about an objective, assumption, initiative, or commitment.	Definition, source, baseline, target, quality, and observation time.

Strategic integrity

- An objective without an accountable owner, horizon, and measure is incomplete.
- An assumption without evidence, monitoring indicators, and invalidation conditions is uncontrolled.
- An initiative without an objective path is orphaned work and enters governance review.
- A recommendation without an objective, option, consequence, and authority path cannot become a decision package.
- A commitment remains linked to the decision context and later outcomes for Decision Replay.

ARCHITECTURE RULE
The Strategy Graph does not score strategy by hiding judgment inside a composite metric. Every roll-up remains decomposable to objectives, evidence, assumptions, dependencies, risks, opportunities, commitments, and outcomes.

CONSTITUTIONAL INHERITANCE / C-009 · C-017 · C-024 · C-026 · C-034

20 Temporal Truth and Truth-State Separation

The Eye must reconstruct what the world was believed to be, what the platform knew, what a person decided, and what later became known—without hindsight contamination.

BITEMPORAL STORAGE PLUS EVENT AND OBSERVATION TIME

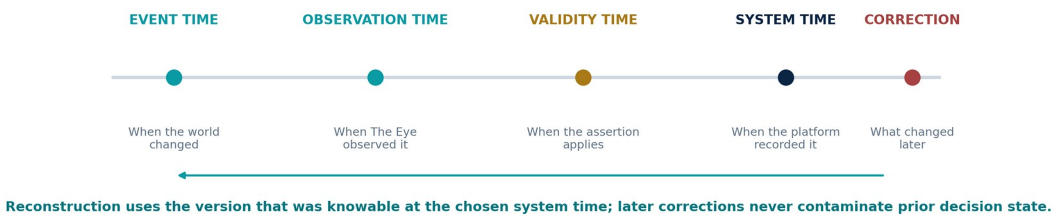

Figure 13. Four-time temporal truth model with later correction preserved separately.

Bitemporal architecture

Material objects carry valid time and system time. Valid time describes when a statement, relationship, state, policy, or assumption applies in the world. System time describes when The Eye accepted, changed, or withdrew the corresponding object version. Observation time is recorded separately because a source may report an earlier event much later.

Current views are projections over immutable version history. Corrections, late evidence, retractions, merges, splits, and adjudications create new versions and linked events. They do not erase the state that informed an earlier forecast, simulation, recommendation, or decision.

Canonical truth states

TRUTH STATE	MEANING	PROMOTION CONSTRAINT
Observed	Directly preserved from an identified source or instrument.	Authenticity and custody do not imply substantive truth.
Claimed	Asserted by a source, person, document, or system.	Must retain source attribution and contradicting claims.
Inferred	Derived by a declared analytical, statistical, model, or agent method.	Requires method, evidence, uncertainty, version, and evaluation.
Assessed	A versioned analytical judgment made or accepted by an accountable human role.	Requires rationale, dissent, evidence coverage, and review state.
Decided	A proposition or option incorporated into a recorded human decision.	Applies within the scope and time of that decision; it is not universal fact.
Synthetic	Created for a scenario, simulation, test, or training purpose.	Must remain isolated from observed state unless explicitly reconciled.
Withdrawn / superseded	No longer decision-active because of correction, expiry, invalidation, or replacement.	History remains available and downstream impact is propagated.

Temporal query modes

- Current effective state for operational use, with freshness and partial-state markers.
- State valid at a declared world time, based on the best current reconstruction.
- State known at a declared system time, preserving the information boundary of an earlier decision.
- Change set between two revisions, including corrections and downstream impact.
- Scenario or simulation state, explicitly isolated from observed and assessed truth.

CONSTITUTIONAL INHERITANCE / C-010 · C-011 · C-019 · C-025 · C-035

21 Trust, Provenance, and Chain of Custody

Trust is constructed from evidence, identity, custody, transformation lineage, quality, policy, explanation, and accountable action. It is not a confidence score attached at the end of a pipeline.

THE CANONICAL LINEAGE NEVER BREAKS AT MODEL, AGENT, OR HUMAN BOUNDARIES

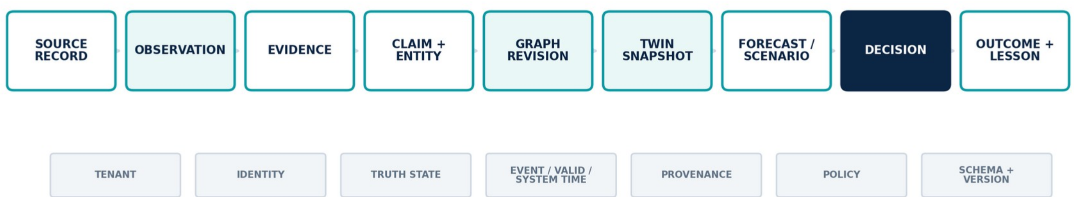

Figure 14. Canonical object lineage from source record to outcome and governed lesson.

Provenance graph

Every material conclusion must support backward explanation and forward impact analysis. Backward explanation resolves the sources, evidence fragments, transformations, graph revisions, memory versions, twin state, models, assumptions, scenarios, simulations, policies, agents, reviewers, and approvals that produced an object. Forward impact identifies active forecasts, warnings, scenarios, simulations, decision packages, briefings, and commitments that depend on an object when it changes.

PROVENANCE EVENT	MINIMUM EVIDENCE
Acquisition	Source identity, authority, time, endpoint, checksum, custody actor, tenant, classification, and raw reference.
Transformation	Input versions, code/model/prompt version, parameters, execution identity, output checksum, evaluation, and exceptions.
Graph or memory commit	Object versions, validation result, ontology and policy revision, actor, causation, and transaction boundary.
Forecast / simulation	Resolved data, twin, scenario, model, environment, seed, constraints, calibration or validation, and run manifest.
Recommendation / decision	Evidence set, alternatives, uncertainty, criteria, rationale, dissent, human authority, policy, and time.
Correction / withdrawal	Affected object, reason, authority, replacement or tombstone, propagation scope, completion, and unresolved consumers.

Evidence integrity

- Canonical evidence is content-addressed or equivalently integrity-protected and immutable by version.
- Derived artifacts link to exact input versions and relevant fragments, not merely to a document title or search result.
- Administrative access and material transformations generate tamper-evident audit evidence.
- Explanation paths are policy-filtered without fabricating continuity where a viewer lacks access.
- Missing, disputed, stale, or withdrawn evidence remains visible in product quality and decision completeness.

TRUST RULE

The platform must be able to say not only why a conclusion was produced, but which conclusions may no longer be safe when an input is corrected or withdrawn.

CONSTITUTIONAL INHERITANCE / C-005 · C-012 · C-035 · C-036 · C-049

22 Identity, Access, and Policy Enforcement

Every human, device, workload, source, connector, agent, model, package, tenant, and customer domain is an authenticated principal operating under continuously evaluated policy.

Authorization decision

DIMENSION	REQUIRED CONTEXT
Principal	Verified identity, role, organization, workload or agent owner, assurance level, session, and device posture.
Purpose	Declared business or mission purpose, legal basis, requested outcome, and allowed secondary use.
Object	Tenant, domain, owner, classification, residency, truth state, lifecycle, sensitivity, and subject rights.
Action	Read, search, infer, transform, export, approve, administer, execute, delete, or delegate.
Environment	Deployment cell, network, region, time, threat state, data boundary, and operational mode.
Obligation	Redaction, watermarking, human review, logging, rate, retention, notification, or dual control.

Enforcement points

Policy enforcement occurs at ingress, object access, search, graph traversal, context assembly, model invocation, agent tool use, workflow transition, approval, export, external execution, administration, and deletion. A front-end visibility rule is not an access control. Service and data boundaries independently enforce the same decision.

- Deny by default and grant the smallest action, object scope, purpose, duration, and environment required.
- Separate strategic authority, content stewardship, platform administration, security operations, and audit review.
- Record policy revision and decision result for material actions so later replay is meaningful.
- Re-evaluate long-running workflows and agent executions when identity, policy, classification, or threat state changes.
- Emergency access is time-bound, conspicuous, independently reviewed, and never erases the audit path.

CONSTITUTIONAL INHERITANCE / C-028 · C-035 · C-036 · C-037 · C-039 · C-040

23 Multi-Agent Orchestration

Agents are bounded execution roles inside governed workflows. They extend analytical reach but do not form an autonomous authority plane and cannot acquire undeclared tools, data, purpose, or decision rights.

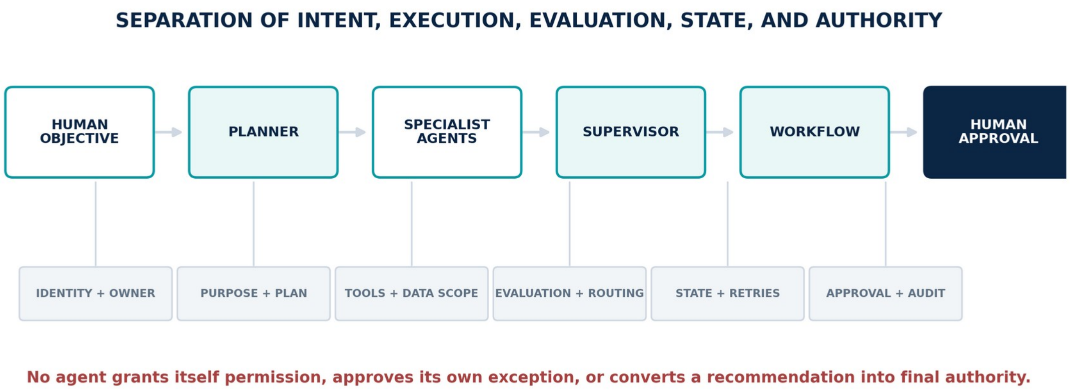

Figure 15. Multi-agent orchestration separating human intent, planning, specialist execution, supervision, workflow, and approval.

Agent execution contract

CONTRACT FIELD	REQUIRED ARCHITECTURE BEHAVIOR
Identity and owner	Unique agent and version, accountable human or organizational owner, tenant, and deployment domain.
Purpose and plan	Declared objective, permitted decomposition, expected products, stop conditions, and escalation path.
Capability grant	Explicit tools, data classes, object scopes, models, networks, side effects, and time window.
Budgets	Token, compute, cost, latency, recursion, concurrency, retries, and external-call limits.
Evidence and state	Immutable inputs, working-state boundary, source citations, produced objects, and truth-state constraints.
Evaluation	Pre-release tests, runtime monitors, product-specific quality checks, policy checks, and post-run assessment.
Supervision	Planner, supervisor, workflow, human review thresholds, interruption, quarantine, rollback, and audit.

Agent families and composition

Observation, Crawler, Search, Collection, Cleaning, Classification, Summarization, NER, Relationship, Knowledge Graph, Reasoning, Prediction, Scenario, Simulation, Decision, Risk, Opportunity, Competitor, Supply Chain, Geopolitical, Technology, Cyber, Financial, Executive Briefing, Reporting, Governance, Learning, Planner, Supervisor, and Workflow agents are capability families. Each deployment instantiates only approved versions and scopes.

The Planner decomposes work; the Workflow service persists and constrains state; the Supervisor evaluates progress, policy, product quality, and budget; specialized agents execute bounded tasks; human gates authorize consequential promotion or action. No agent can approve its own expanded authority or silently modify its governing prompt, tools, policy, model, evaluation, or release state.

Containment and failure behavior

- Treat retrieved content and external instructions as untrusted data, not authority.
- Use capability-scoped tool brokers; do not expose ambient credentials or unrestricted network access.
- Persist intermediate claims and evidence links so a supervisor can challenge, resume, or reproduce the work.
- Stop, abstain, or route to a human when evidence, policy, evaluation, budget, or authority is insufficient.
- Quarantine anomalous executions and invalidate dependent outputs when a model, package, source, or agent version is withdrawn.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-028 · C-029 · C-031 · C-045

24 Model Gateway and Model Pluralism

The architecture separates strategic product contracts from model providers. Models are governed, evaluated, replaceable computational dependencies—not authorities and not hidden system boundaries.

Model gateway responsibilities

- Resolve an approved model and version by task, domain, risk, deployment, data boundary, language, cost, latency, and quality policy.
- Apply input minimization, classification, routing, encryption, regional and provider restrictions, and retention controls.
- Bind prompts, tools, retrieval context, parameters, safety policy, and output schema to a versioned invocation plan.
- Capture invocation lineage, usage, latency, evaluation, failure, abstention, and provider-specific evidence without exposing sensitive content to global telemetry.
- Support deterministic, statistical, symbolic, optimization, domain, open-weight, customer-provided, and foundation models behind one governed contract.

Selection and fallback

CONDITION	REQUIRED BEHAVIOR
Primary unavailable	Use only a pre-approved equivalent route; otherwise abstain or queue. Never silently weaken residency, privacy, or quality policy.
Quality threshold missed	Try a declared ensemble, deterministic method, alternate model, or human review path; retain disagreement.
Drift or incident	Quarantine the affected version, identify dependent objects, activate rollback, and initiate re-evaluation.
On-Premise / disconnected	Resolve locally available approved models and make reduced capability explicit in product state.
Provider change	Require compatibility, regression, safety, cost, latency, and explanation evaluation before promotion.

ARCHITECTURE RULE

A model substitution is not equivalent merely because its interface accepts the same payload. It must preserve task semantics, data controls, evaluation thresholds, explanation evidence, and failure behavior.

CONSTITUTIONAL INHERITANCE / C-005 · C-030 · C-031 · C-035 · C-041 · C-045

25 Workflow, State, and Human Gates

Long-running strategic work is coordinated as durable, observable state machines with explicit ownership, human tasks, retries, compensation, deadlines, evidence, and terminal conditions.

Workflow state model

STATE CLASS	ARCHITECTURE MEANING
Draft / planned	Purpose, owner, scope, versions, expected products, controls, and budgets are declared before work begins.
Active / waiting	Execution is in progress or blocked on time, data, policy, review, approval, external response, or resource availability.
Challenged / remediating	A quality, evidence, policy, model, or human challenge requires revision while preserving earlier state.
Approved / committed	A named authority has accepted the package and activated obligations or downstream action.
Rejected / cancelled	Work ended without promotion; reason, actor, affected outputs, and cleanup or compensation remain recorded.
Completed / archived	Required products, evidence, obligations, notifications, and evaluation are complete and retained under policy.

Human gate contract

- The gate identifies accountable role, delegated authority, required independence, information package, permitted actions, deadline, and escalation.
- Approval records the exact object versions, policy revision, rationale, conditions, dissent, overrides, and authenticated actor.
- A later correction reopens or invalidates affected work through a new transition; it never rewrites the earlier approval.
- No timeout converts silence into approval for a consequential decision or external side effect.
- Workflow definitions are versioned release artifacts; in-flight instances retain or deliberately migrate their semantics.

Reliability semantics

The architecture assumes retries, duplicates, partial failures, delayed events, and human latency. Commands use stable idempotency keys where reissue is possible. Transitions commit with causation and audit evidence. External side effects use a governed outbox, callback, or reconciliation pattern so The Eye can prove intent, execution, rejection, and residual uncertainty.

CONSTITUTIONAL INHERITANCE / C-004 · C-025 · C-029 · C-036 · C-043 · C-049

26 Evaluation and Continuous Learning

Learning is a governed release process. The Eye measures products and outcomes continuously, proposes changes, tests them, obtains approval, deploys them safely, and preserves rollback.

Evaluation portfolio

OBJECT UNDER EVALUATION	REPRESENTATIVE MEASURES
Sources and collection	Coverage, freshness, authenticity, latency, correction rate, blind spots, and rights compliance.
Extraction and graph	Precision, recall, calibration, identity accuracy, contradiction retention, ontology conformance, and orphan rate.
Retrieval and explanation	Relevance, evidence coverage, permission correctness, citation validity, temporal correctness, and answer completeness.
Forecast and scenario	Proper scoring rules, calibration, sharpness, bias, scenario diversity, coherence, signpost utility, and drift.
Simulation and decision	Reproducibility, stability, sensitivity coverage, decision completeness, alternative coverage, and outcome linkage.
Agents and workflows	Task success, policy compliance, tool correctness, abstention, human override, budget, recovery, and side-effect integrity.
Institutional outcomes	Decision latency, warning precision, commitment delivery, strategic objective movement, lesson adoption, and unintended consequence.

Governed learning loop

An outcome, incident, drift signal, challenge, or evaluation result may create a Lesson Proposal. The proposal identifies evidence, affected components and products, expected benefit, risks, validation plan, owner, approval path, deployment scope, monitoring, and rollback. Approved changes are released through the same configuration, schema, model, agent, workflow, ontology, and policy controls as any other architecture change.

- No interaction is silently converted into training data or shared improvement without authority and policy.
- Offline benchmarks, adversarial tests, replay, shadow evaluation, and controlled production metrics are used together.
- Evaluation sets are versioned, representative, protected from contamination, and segmented by domain and risk.
- Material regressions block release or activate rollback even when aggregate scores improve.
- Institutional lessons remain explainable and reversible; they do not become unreviewed memory or policy.

CONSTITUTIONAL INHERITANCE / C-031 · C-044 · C-045 · C-049 · C-050

27 Search, Retrieval, and Context Assembly

Retrieval is a governed intelligence operation across evidence, memory, graph, twins, decisions, and outcomes. It must preserve permissions, time, truth state, provenance, coverage, and purpose.

Federated retrieval plan

The Context Assembly service translates a user, workflow, model, or agent request into a policy-filtered retrieval plan. The plan may combine lexical and semantic document retrieval, graph traversal, temporal filters, structured queries, twin state, decision history, and source reacquisition. Results are ranked after authorization and carry exact object versions and evidence fragments.

STAGE	CONTRACT
Interpret	Purpose, actor, question, scope, time, truth state, quality, and expected output are explicit.
Authorize	Candidate object access is evaluated before content reaches ranking, model, or agent context.
Retrieve	Search indexes, graph, memory, and structured projections return stable references and revision metadata.
Assemble	Deduplicate, diversify, resolve contradictions, allocate token or payload budget, and preserve coverage gaps.
Explain	Return citations, paths, excluded-result categories, freshness, confidence, partial state, and query revision.

Safety and correctness

- Search indexes are projections; canonical objects remain the authority for content, policy, and version history.
- Security trimming is enforced before model context, not repaired after generation.
- Retrieved instructions are treated as evidence content and cannot expand agent authority.
- Temporal queries distinguish what is currently believed from what was knowable at an earlier decision time.
- A low-coverage result declares insufficiency and may request acquisition or human input instead of fabricating completeness.

CONSTITUTIONAL INHERITANCE / C-005 · C-010 · C-011 · C-016 · C-035 · C-040

28 Marketplace Package Architecture

The Agent Marketplace and Scenario Marketplace distribute governed capability packages without creating an uncontrolled path into customer data, models, networks, workflows, or strategic decisions.

Signed package manifest

MANIFEST AREA	REQUIRED DECLARATION
Identity	Publisher, package, semantic version, signature, provenance, licence, support, and end-of-life.
Purpose	Supported use cases, prohibited uses, expected outputs, target domains, and responsible owner.
Dependencies	Ontology, schema, agent, model, connector, workflow, policy, twin, data class, and runtime compatibility.
Capabilities	Tools, networks, object scopes, external effects, secrets, budgets, and human gates requested.
Quality and safety	Evaluation results, known limitations, threat analysis, red-team evidence, monitoring, and rollback.
Data behavior	Inputs, outputs, retention, training use, telemetry, residency, exports, and deletion semantics.

Lifecycle

- Ingest into quarantine; verify signature, provenance, licence, malware, dependencies, policy, and declared behavior.
- Evaluate with representative fixtures and adversarial tests inside an isolated sandbox.
- Approve for a customer domain with explicit capability grants; installation alone grants no access.
- Monitor package behavior, output quality, dependency health, publisher status, and discovered vulnerabilities.
- Suspend, revoke, or remove a package while preserving affected executions, decisions, and replay evidence.

MARKETPLACE RULE
Portability of capability never implies portability of authority. Every installed package remains subordinate to customer policy, deployment boundaries, human decision rights, and local evaluation.

CONSTITUTIONAL INHERITANCE / C-028 · C-035 · C-041 · C-046 · C-050

29 Proactive Intelligence and Early Warning

The Eye identifies material change without waiting for a query, but it earns attention through consequence, evidence, calibration, ownership, and disciplined closure—not volume.

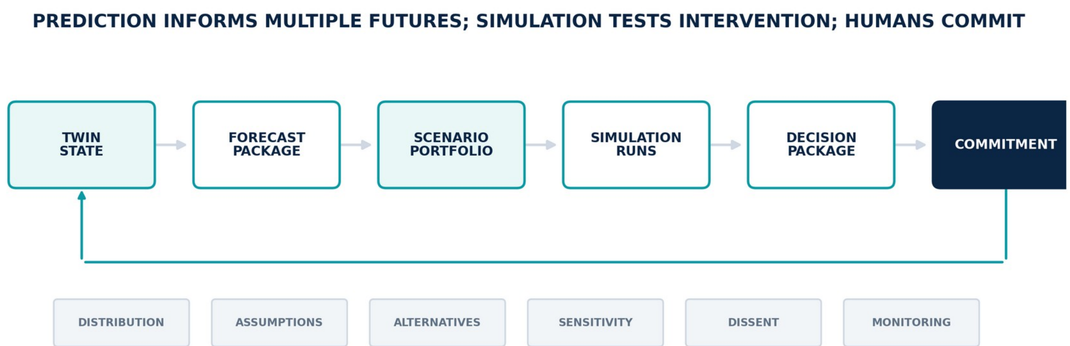

Figure 16. Foresight and decision flow from twin state through forecasts, scenarios, simulations, decisions, and commitments.

Warning product contract

FIELD	REQUIRED CONTENT
Change	What changed, when, where, and relative to which baseline, expectation, threshold, or scenario.
Evidence	Supporting and contradicting evidence, source health, provenance, freshness, and coverage gaps.
Assessment	Confidence, uncertainty, mechanism, weak-signal status, novelty, and alternative explanations.
Consequence	Affected objectives, assets, obligations, twins, scenarios, forecasts, commitments, and response window.
Action	Accountable owner, available response paths, information needs, decision requirement, and escalation policy.
Lifecycle	Deduplication key, acknowledgement, suppression, update, escalation, resolution, outcome, and evaluation.

Weak signal architecture

Weak Signal Detection combines novelty, acceleration, relationship change, diffusion, source diversity, contradiction, anomaly, emerging vocabulary, graph centrality, and strategic relevance. A weak signal is not promoted because it is unusual. It is promoted when the platform can show a plausible mechanism connecting the change to an objective, exposure, capability, scenario, or decision window.

Attention discipline

- Deduplicate related signals into one evolving intelligence product.
- Prioritize consequence, urgency, confidence, reversibility, and information value under role-specific attention policy.
- Suppress through governed rules with visible rationale, duration, owner, and audit—not through silent model filtering.
- Track missed warnings, false alarms, timeliness, acknowledgement, action, and outcome by signal class.
- Separate risk and opportunity so favorable change receives the same analytical discipline as threat.

CONSTITUTIONAL INHERITANCE / C-005 · C-032 · C-033 · C-034 · C-047 · C-049

PART IV

Runtime Intelligence Flows

End-to-end flows that preserve identity, time, provenance, policy, and human authority across the strategic loop.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

30 Observe-to-Evidence Flow

The acquisition path turns an authorized source interaction into immutable, tenant-bound, policy-classified evidence with measured coverage and chain of custody.

Trigger and admission

A scheduled collection, source event, subscription message, user-authorized retrieval, internal-system change, or sensor observation enters the Source Contract Service.

Canonical sequence

STEP	ACTION	ARCHITECTURE BEHAVIOR	COMMITTED PRODUCT
30.1	Authorize acquisition	Resolve source contract, purpose, rights, tenant, credential reference, rate, residency, and collection window.	Source Contract version and Policy Decision
30.2	Acquire and timestamp	Capture payload and transport metadata; assign event, observation, and receipt times plus stable correlation.	Observation Record candidate
30.3	Inspect and quarantine	Verify authenticity where available; scan, classify, and isolate unsafe or malformed content.	Qualified or quarantined payload
30.4	Preserve raw evidence	Write immutable content, checksum, custody events, source reference, tenant, classification, and retention state.	Evidence Object
30.5	Measure source state	Update freshness, latency, completeness, authenticity, collection success, coverage, and blind spots.	Source health and coverage revision
30.6	Publish observation	Commit the ObservationRecorded event only after evidence identity and custody are durable.	Replayable domain event

Control points

- No collection without an active source contract and customer purpose.
- Unsafe payloads never enter transformation context unmarked.
- Evidence preservation precedes derived intelligence.
- Failures alter source health and dependent product quality.

Failure and recovery conditions

FAILURE CONDITION	REQUIRED RESPONSE
Credential or authority failure	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Source unavailable or rate-limited	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Authenticity uncertain	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Malware or prompt-injection content	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Schema drift	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Residency or classification conflict	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

Products and observability

- Evidence Object and Observation Record
- Custody and audit evidence
- Source health and coverage state
- Quarantine or exception workflow

SIGNAL FAMILY	REQUIRED FLOW EVIDENCE
Execution state	Step duration, end-to-end age, queue and wait time, retry, compensation, completion, and unresolved dependency.
Trust state	Policy decisions, source and object versions, transformation, model and agent lineage, review, approval, and audit continuity.
Product state	Completeness, freshness, coverage, uncertainty, invalidation, affected objectives and consumers, and decision-active status.
Recovery state	Last durable transition, idempotency and causation, checkpoints, external effects, replay position, and accountable owner.

CONSTITUTIONAL INHERITANCE / C-012 · C-013 · C-014 · C-035 · C-039

31 Evidence-to-Understanding Flow

The interpretation path converts preserved evidence into structured, contestable intelligence and governed graph change without collapsing claims, inferences, and assessments into one truth state.

Trigger and admission

An ObservationRecorded event or authorized reprocessing request selects an approved transformation plan for the evidence type, language, domain, and risk class.

Canonical sequence

STEP	ACTION	ARCHITECTURE BEHAVIOR	COMMITTED PRODUCT
31.1	Resolve transformation	Bind parser, cleaning, classification, extraction, models, prompts, ontology, policy, and evaluation versions.	Transformation manifest
31.2	Normalize evidence	Extract structure, text, metadata, units, language, and fragments while retaining raw coordinates and checksums.	Normalized evidence view
31.3	Extract intelligence	Propose entities, relationships, events, claims, attributes, classifications, and summaries with confidence and lineage.	Candidate intelligence objects
31.4	Corroborate and challenge	Retrieve related evidence; detect contradiction, duplication, novelty, missing context, and alternative interpretations.	Evidence and contradiction set
31.5	Adjudicate promotion	Apply deterministic validation, thresholds, stewardship, and human review appropriate to impact and ambiguity.	Approved, rejected, or pending candidates
31.6	Commit understanding	Create memory versions and atomic graph revision; publish affected identities, paths, twins, and downstream products.	Claims, Events, Assessment, GraphRevision

Control points

- Every derived object cites exact evidence fragments and method versions.
- Contradictory claims coexist until governed adjudication.
- Model output remains inferred unless promoted through an authorized process.
- High-impact or low-confidence promotion requires human review.

Failure and recovery conditions

FAILURE CONDITION	REQUIRED RESPONSE
Unsupported format or language	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Extraction confidence below threshold	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Ontology incompatibility	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Entity ambiguity	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Prompt injection or unsafe content	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Required reviewer unavailable	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

Products and observability

- Structured intelligence objects
- Graph and memory revisions
- Contradiction and review state
- Evaluation and transformation lineage

SIGNAL FAMILY	REQUIRED FLOW EVIDENCE
Execution state	Step duration, end-to-end age, queue and wait time, retry, compensation, completion, and unresolved dependency.
Trust state	Policy decisions, source and object versions, transformation, model and agent lineage, review, approval, and audit continuity.
Product state	Completeness, freshness, coverage, uncertainty, invalidation, affected objectives and consumers, and decision-active status.
Recovery state	Last durable transition, idempotency and causation, checkpoints, external effects, replay position, and accountable owner.

CONSTITUTIONAL INHERITANCE / C-005 · C-009 · C-010 · C-012 · C-029 · C-045

32 Correction and Withdrawal Propagation

Corrections are first-class system events. The architecture preserves the earlier record, versions the corrected state, identifies every decision-active dependency, and proves propagation or unresolved impact.

Trigger and admission

A source correction, legal request, stewardship decision, model incident, identity merge or split, ontology correction, data-quality finding, or human challenge changes the fitness of an existing object.

Canonical sequence

STEP	ACTION	ARCHITECTURE BEHAVIOR	COMMITTED PRODUCT
32.1	Authenticate correction	Verify authority, reason, affected version, scope, legal or policy basis, and supporting evidence.	Correction case
32.2	Version canonical state	Create replacement, withdrawal, tombstone, merge, split, or adjudication object; never overwrite history.	New object version and causation event
32.3	Compute impact	Traverse provenance and subscriptions to find graph, memory, twins, forecasts, warnings, scenarios, simulations, decisions, briefings, and commitments.	Impact set
32.4	Invalidate or mark	Change decision-active status, freshness, confidence, or completeness according to policy and materiality.	Affected product revisions
32.5	Recompute or review	Re-run eligible deterministic products; route high-impact or committed products to human owners.	Replacement products or review tasks
32.6	Close propagation	Record reached consumers, failed deliveries, residual exposure, owner acknowledgement, and final audit evidence.	Propagation completion evidence

Control points

- Historical decisions retain their original information boundary.
- Current projections exclude withdrawn material according to policy.
- No automatic rewrite of a human decision or commitment.
- Unresolved downstream impact remains visible until owned and closed.

Failure and recovery conditions

FAILURE CONDITION	REQUIRED RESPONSE
Correction authority disputed	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Provenance path incomplete	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Consumer unavailable	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Committed action already executed	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Legal hold conflicts with deletion	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

FAILURE CONDITION	REQUIRED RESPONSE
Recomputation changes recommendation materially	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

Products and observability

- Corrected canonical object
- Impact and invalidation graph
- Recomputation or review workflow
- Propagation audit evidence

SIGNAL FAMILY	REQUIRED FLOW EVIDENCE
Execution state	Step duration, end-to-end age, queue and wait time, retry, compensation, completion, and unresolved dependency.
Trust state	Policy decisions, source and object versions, transformation, model and agent lineage, review, approval, and audit continuity.
Product state	Completeness, freshness, coverage, uncertainty, invalidation, affected objectives and consumers, and decision-active status.
Recovery state	Last durable transition, idempotency and causation, checkpoints, external effects, replay position, and accountable owner.

CONSTITUTIONAL INHERITANCE / C-011 · C-012 · C-016 · C-025 · C-036 · C-043

33 Graph-to-Twin Reconciliation

Digital Twins remain grounded by reconciling new evidence and graph revisions against declared twin boundaries, mappings, state estimators, constraints, assumptions, and human stewardship.

Trigger and admission

A GraphChanged event, telemetry update, internal-system change, correction, ontology revision, or scheduled reconciliation affects an entity or relationship mapped to a Digital Twin.

Canonical sequence

STEP	ACTION	ARCHITECTURE BEHAVIOR	COMMITTED PRODUCT
33.1	Resolve affected twins	Use graph subscriptions and mapping versions to identify affected twin instances and state variables.	Twin impact set
33.2	Qualify input	Evaluate source health, truth state, time, confidence, policy, mapping, units, and expected cadence.	Qualified state observation
33.3	Estimate candidate state	Apply declared estimator or model without mutating the active snapshot; retain uncertainty and disagreement.	Candidate twin delta
33.4	Validate constraints	Check topology, conservation, ranges, business rules, temporal ordering, and cross-twin dependencies.	Validation result
33.5	Review material changes	Route ambiguous, discontinuous, policy-sensitive, or high-consequence changes to the accountable twin owner.	Approval or challenge
33.6	Publish snapshot	Create a new versioned snapshot and issue change events to forecasts, scenarios, simulations, risks, and warnings.	Twin Snapshot and change set

Control points

- Observed state is separated from estimated and simulated state.
- Mappings and estimators are versioned dependencies.
- A simulation cannot write into the active observed twin.
- Material gaps and stale state remain visible.

Failure and recovery conditions

FAILURE CONDITION	REQUIRED RESPONSE
Mapping no longer valid	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Conflicting observations	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Constraint violation	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Telemetry gap	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Estimator drift	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Cross-twin dependency unavailable	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

Products and observability

- Versioned Twin Snapshot
- State delta and uncertainty
- Quality and gap state
- Downstream subscriptions

SIGNAL FAMILY	REQUIRED FLOW EVIDENCE
Execution state	Step duration, end-to-end age, queue and wait time, retry, compensation, completion, and unresolved dependency.
Trust state	Policy decisions, source and object versions, transformation, model and agent lineage, review, approval, and audit continuity.
Product state	Completeness, freshness, coverage, uncertainty, invalidation, affected objectives and consumers, and decision-active status.
Recovery state	Last durable transition, idempotency and causation, checkpoints, external effects, replay position, and accountable owner.

CONSTITUTIONAL INHERITANCE / C-009 · C-011 · C-018 · C-019 · C-035

34 Forecast Production Flow

A forecast is a governed probability product bound to a precise target, horizon, information boundary, data, twin, model, calibration, assumptions, and explanation.

Trigger and admission

A scheduled horizon update, material evidence change, twin revision, scenario signpost, decision request, or analyst command opens a Forecast Run.

Canonical sequence

STEP	ACTION	ARCHITECTURE BEHAVIOR	COMMITTED PRODUCT
34.1	Declare forecast target	Define outcome, population or system, issue time, target window, horizons, units, owner, and decision use.	Forecast specification
34.2	Freeze information set	Resolve evidence, graph revision, twin snapshot, features, assumptions, policy, and known coverage gaps.	Information boundary manifest
34.3	Select approved methods	Resolve eligible deterministic, statistical, domain, and machine-learning models under deployment and risk policy.	Model and ensemble plan
34.4	Generate distributions	Produce horizon-specific distributions, quantiles, intervals, drivers, sensitivities, and model disagreement.	Forecast candidates
34.5	Calibrate and challenge	Apply backtests, proper scoring rules, drift checks, bias tests, expert review, and alternative assumptions.	Evaluation and challenge record
34.6	Publish package	Commit approved forecast with failure conditions, monitoring, update triggers, explanation, and downstream links.	Forecast Package

Control points

- All six canonical horizons are supported where the target is meaningful.
- Point estimates cannot replace probability distributions.
- Model disagreement and weak coverage remain visible.
- Human judgment is labeled and versioned separately from model output.

Failure and recovery conditions

FAILURE CONDITION	REQUIRED RESPONSE
Target not forecastable	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Insufficient history or coverage	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Model drift	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Calibration failure	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Horizon mismatch	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Policy or deployment prevents approved method	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

Products and observability

- Forecast Package
- Calibration and drift state
- Driver and sensitivity explanation
- Scenario and decision subscriptions

SIGNAL FAMILY	REQUIRED FLOW EVIDENCE
Execution state	Step duration, end-to-end age, queue and wait time, retry, compensation, completion, and unresolved dependency.
Trust state	Policy decisions, source and object versions, transformation, model and agent lineage, review, approval, and audit continuity.
Product state	Completeness, freshness, coverage, uncertainty, invalidation, affected objectives and consumers, and decision-active status.
Recovery state	Last durable transition, idempotency and causation, checkpoints, external effects, replay position, and accountable owner.

CONSTITUTIONAL INHERITANCE / C-005 · C-020 · C-021 · C-030 · C-045

35 Scenario Branch Lifecycle

Scenario Intelligence maintains several coherent futures at once, monitors the evidence that would distinguish them, and preserves retired branches for learning and Decision Replay.

Trigger and admission

A strategic question, risk or opportunity, forecast shift, weak signal, executive request, planning cycle, or imported governed package creates or updates a scenario portfolio.

Canonical sequence

STEP	ACTION	ARCHITECTURE BEHAVIOR	COMMITTED PRODUCT
35.1	Establish baseline	Freeze current evidence, graph revision, twin state, forecasts, objectives, and planning horizon.	Scenario baseline
35.2	Declare drivers and actors	Model mechanisms, dependencies, constraints, uncertainties, agency, and exogenous conditions.	Driver and actor model
35.3	Create branches	Generate baseline, upside, downside, disruption, and domain-specific branches with explicit divergence logic.	Scenario Branches
35.4	Validate coherence	Check internal consistency, temporal ordering, assumptions, graph dependencies, forecast relationships, and prohibited contradictions.	Coherence assessment
35.5	Define signposts	Attach observable indicators, thresholds, source contracts, owners, confidence, and update cadence.	Monitoring contract
35.6	Review portfolio	Compare branch relevance, consequences, option robustness, missing branches, and retirement or simulation candidates.	Governed scenario portfolio

Control points

- No single branch is presented as the future.
- Assumptions and branch divergence are explicit.
- Imported scenarios execute without ambient customer access.
- Retirement preserves version history and decision links.

Failure and recovery conditions

FAILURE CONDITION	REQUIRED RESPONSE
Branch incoherent	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Critical assumption invalidated	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Signpost unavailable	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Branches converge or fail to cover material uncertainty	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Imported package incompatible	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

FAILURE CONDITION	REQUIRED RESPONSE
Owner misses review cadence	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

Products and observability

- Scenario portfolio
- Assumption register
- Indicators and signposts
- Simulation candidates and decision context

SIGNAL FAMILY	REQUIRED FLOW EVIDENCE
Execution state	Step duration, end-to-end age, queue and wait time, retry, compensation, completion, and unresolved dependency.
Trust state	Policy decisions, source and object versions, transformation, model and agent lineage, review, approval, and audit continuity.
Product state	Completeness, freshness, coverage, uncertainty, invalidation, affected objectives and consumers, and decision-active status.
Recovery state	Last durable transition, idempotency and causation, checkpoints, external effects, replay position, and accountable owner.

CONSTITUTIONAL INHERITANCE / C-022 · C-023 · C-032 · C-033 · C-046

36 Simulation Run Lifecycle

Simulation tests interventions before commitment through reproducible experiments that never mutate observed truth or hide unresolved assumptions.

Trigger and admission

An approved analyst, agent, workflow, or decision package submits an experiment against a declared twin snapshot and scenario branch.

Canonical sequence

STEP	ACTION	ARCHITECTURE BEHAVIOR	COMMITTED PRODUCT
36.1	Define experiment	Declare question, intervention, constraints, measures, expected outputs, budgets, validation, and owner.	Experiment specification
36.2	Resolve immutable inputs	Bind twin, scenario, graph, model, solver, policy, data, configuration, and environment versions.	Run manifest
36.3	Validate feasibility	Check required state, constraints, capability, policy, capacity, determinism, and stop conditions.	Admission decision
36.4	Execute and checkpoint	Run isolated work with stable seeds where applicable, resource controls, progress events, and recoverable checkpoints.	Simulation execution trace
36.5	Analyze impacts	Compute direct, indirect, second-order, distributional, temporal, sensitivity, and robustness results.	Impact and sensitivity set
36.6	Validate and publish	Challenge assumptions and stability; attach reproducibility evidence; release or invalidate the run.	Simulation Run package

Control points

- Exact input and execution versions are resolvable.
- Synthetic state remains isolated from observed and assessed state.
- Material results include sensitivity and validation.
- Invalidation propagates to active decision packages.

Failure and recovery conditions

FAILURE CONDITION	REQUIRED RESPONSE
Input version missing	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Constraint unsatisfied	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Solver unstable	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Budget or timeout exceeded	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Partial run	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Validation or reproducibility failure	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

Products and observability

- Simulation Run
- Impact, sensitivity, and uncertainty
- Reproducibility package
- Challenge or invalidation record

SIGNAL FAMILY	REQUIRED FLOW EVIDENCE
Execution state	Step duration, end-to-end age, queue and wait time, retry, compensation, completion, and unresolved dependency.
Trust state	Policy decisions, source and object versions, transformation, model and agent lineage, review, approval, and audit continuity.
Product state	Completeness, freshness, coverage, uncertainty, invalidation, affected objectives and consumers, and decision-active status.
Recovery state	Last durable transition, idempotency and causation, checkpoints, external effects, replay position, and accountable owner.

CONSTITUTIONAL INHERITANCE / C-018 · C-019 · C-023 · C-035 · C-043 · C-045

37 Decision Package and Approval Flow

Decision Intelligence assembles a complete choice environment and transfers final authority to an authenticated human decision-maker through a durable approval workflow.

Trigger and admission

A named authority, delegated workflow, material warning, commitment review, scenario threshold, or strategic cadence opens a decision with declared scope and deadline.

Canonical sequence

STEP	ACTION	ARCHITECTURE BEHAVIOR	COMMITTED PRODUCT
37.1	Frame the decision	Declare question, owner, final authority, objectives, stakeholders, constraints, obligations, timing, and reversibility.	Decision definition
37.2	Develop viable options	Include status quo, defer, seek information, and other realistic alternatives; identify dependencies and execution readiness.	Option set
37.3	Assemble intelligence	Resolve evidence, graph, memory, twin, forecasts, scenarios, simulations, risks, opportunities, and prior decisions.	Information boundary
37.4	Compare and recommend	Apply inspectable criteria; show trade-offs, uncertainty, dissent, sensitivity, robustness, value of information, and failure conditions.	Decision Package revision
37.5	Review and decide	Named humans approve, reject, defer, request information, override, or condition the recommendation under separation-of-duties policy.	Final Decision Record
37.6	Create commitments	Bind approved option to owners, resources, obligations, milestones, indicators, review cadence, execution interfaces, and closure criteria.	Commitments and monitoring contracts

Control points

- AI and agents cannot be the final authority.
- Alternatives, dissent, uncertainty, and missing evidence remain visible.
- Approval binds exact package and policy versions.
- External execution requires a separately governed commitment and interface.

Failure and recovery conditions

FAILURE CONDITION	REQUIRED RESPONSE
No named authority	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Mandatory evidence incomplete	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Option not executable	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Separation-of-duties conflict	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Policy denial	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

FAILURE CONDITION	REQUIRED RESPONSE
Material input changes during review	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

Products and observability

- Decision Package
- Human decision and rationale
- Commitments and obligations
- Monitoring and replay snapshot

SIGNAL FAMILY	REQUIRED FLOW EVIDENCE
Execution state	Step duration, end-to-end age, queue and wait time, retry, compensation, completion, and unresolved dependency.
Trust state	Policy decisions, source and object versions, transformation, model and agent lineage, review, approval, and audit continuity.
Product state	Completeness, freshness, coverage, uncertainty, invalidation, affected objectives and consumers, and decision-active status.
Recovery state	Last durable transition, idempotency and causation, checkpoints, external effects, replay position, and accountable owner.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-024 · C-025 · C-026 · C-036 · C-050

38 Decision Replay and Outcome Learning

Decision Replay reconstructs the information, alternatives, assumptions, models, authority, rationale, commitments, and later outcomes of a decision so the institution can learn without rewriting history.

Trigger and admission

A scheduled review, outcome observation, commitment exception, warning, correction, incident, audit, leadership request, or model evaluation opens a replay case.

Canonical sequence

STEP	ACTION	ARCHITECTURE BEHAVIOR	COMMITTED PRODUCT
38.1	Restore decision-time state	Resolve the exact evidence, graph, memory, twins, forecasts, scenarios, simulations, package, policy, and actor authority known then.	Historical information boundary
38.2	Reconstruct reasoning	Present options, criteria, recommendation, dissent, assumptions, uncertainty, approvals, overrides, and rationale without later facts.	Decision narrative and trace
38.3	Connect execution	Link commitments, resources, operational actions, deviations, external conditions, and governance events.	Execution timeline
38.4	Assess outcomes	Compare intended and observed effects, leading and lagging measures, side effects, counterfactual limitations, and attribution confidence.	Outcome evaluation
38.5	Identify lessons	Separate process, evidence, model, forecast, scenario, simulation, governance, execution, and environmental causes.	Lesson Proposals
38.6	Govern change	Test, approve, release, monitor, and retain rollback for accepted changes to memory, models, agents, policy, workflow, ontology, or practice.	Released learning and audit evidence

Control points

- Replay preserves what was knowable at the time.
- Outcome evaluation separates decision quality from luck where evidence allows.
- Lessons remain proposals until governance approves them.
- No evaluation silently trains or changes production behavior.

Failure and recovery conditions

FAILURE CONDITION	REQUIRED RESPONSE
Historical object unavailable	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Execution occurred outside governed interfaces	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Outcome attribution weak	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Policy or identity revision unresolved	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.
Counterfactual impossible	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

FAILURE CONDITION	REQUIRED RESPONSE
Lesson lacks owner or validation	Expose the failure state, preserve causation and partial work, apply the owning layer's degraded behavior, and route retry, compensation, challenge, or human review under policy.

Products and observability

- Replay package
- Outcome and attribution assessment
- Lesson Proposals
- Governed improvement or retained dissent

SIGNAL FAMILY	REQUIRED FLOW EVIDENCE
Execution state	Step duration, end-to-end age, queue and wait time, retry, compensation, completion, and unresolved dependency.
Trust state	Policy decisions, source and object versions, transformation, model and agent lineage, review, approval, and audit continuity.
Product state	Completeness, freshness, coverage, uncertainty, invalidation, affected objectives and consumers, and decision-active status.
Recovery state	Last durable transition, idempotency and causation, checkpoints, external effects, replay position, and accountable owner.

CONSTITUTIONAL INHERITANCE / C-011 · C-015 · C-025 · C-031 · C-044 · C-045

PART V

Deployment and Enterprise Operation

Deployment parity, sovereignty, security, resilience, observability, lifecycle, interoperability, and exit.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

39 Deployment Model and Semantic Parity

SaaS, Private Cloud, and On-Premise are operating envelopes for one product. They may differ in physical topology, managed-service choice, capacity, connectivity, key custody, and operational responsibility; they must not differ in strategic meaning.

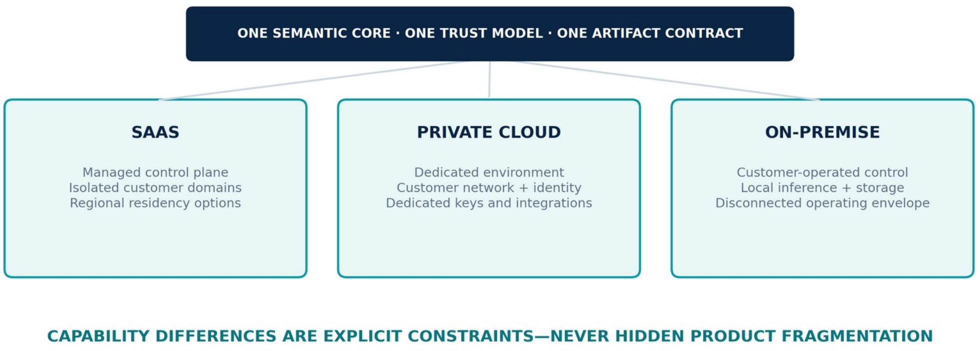

Figure 17. Deployment semantic parity across SaaS, Private Cloud, and On-Premise.

Deployment invariants

- The ten layers, canonical objects, truth states, temporal model, provenance, decision rights, audit semantics, and core workflows remain identical.
- APIs, events, package manifests, ontology contracts, policy decisions, export formats, and conformance fixtures remain portable across modes.
- Capability availability may be constrained by disconnected operation, local models, capacity, or customer policy, but reduced capability is explicit and never presented as equivalent output.
- Customer data, keys, models, administration, telemetry, support, and update boundaries are declared per deployment and independently testable.
- Promotion between environments resolves the same signed artifacts and configuration intent under environment-specific policy.

Logical deployment units

UNIT	RESPONSIBILITY
Customer Intelligence Domain	Isolation boundary for customer evidence, memory, graph, twins, models, agents, workflows, decisions, policies, and audit.
Intelligence Cell	Scalable failure and data locality unit hosting a bounded set of tenant or domain workloads.
Shared Control Services	Versioned identity federation, package distribution, policy templates, licensing, support, and operational metadata within declared boundaries.
Edge / Collection Zone	Controlled acquisition near sources, sensors, restricted networks, or low-connectivity environments with store-and-forward custody.
Execution Gateway	Governed egress for notifications, exports, callbacks, enterprise actions, and other approved external side effects.
Recovery Domain	Independent backups, replicated state, keys, manifests, runbooks, and restore capability aligned to sovereignty policy.

CONSTITUTIONAL INHERITANCE / C-006 · C-037 · C-038 · C-041 · C-042 · C-043

40 SaaS Operating Envelope

The SaaS mode provides a vendor-operated, cell-based service with strong tenant isolation, regional control, customer-visible service boundaries, and independently recoverable failure domains.

Cell architecture

A SaaS cell contains the customer-domain data and execution services required for the canonical product while limiting blast radius, operational contention, and regional scope. A routing and admission layer selects the authorized cell after identity, tenant, region, and service-state checks. Global functions hold only the metadata required to locate, license, update, and operate cells under policy.

- No request trusts a tenant identifier supplied only by the client; identity and routing bind it independently.
- High-risk administrative actions use privileged access management, just-in-time elevation, approval, recording, and customer-visible audit where contracted.
- Cross-cell data movement is denied unless a governed federation, migration, recovery, or customer export workflow authorizes it.
- Regional failover respects residency and key policy; availability does not override sovereignty.
- Noisy-neighbor controls, workload quotas, agent budgets, and admission policies protect strategic workloads under shared operation.

Responsibility boundary

CONTROL AREA	PLATFORM RESPONSIBILITY	CUSTOMER RESPONSIBILITY
Service operation	Cell infrastructure, platform software, monitoring, recovery, vulnerability and release operations.	Tenant configuration, authorized use, business continuity integration, and user governance.
Identity	Federation, workload identity, policy enforcement, privileged operations.	Identity source integrity, role ownership, joiner/mover/leaver process, and authority delegation.
Data	Isolation, encryption, backups, retention enforcement, export and deletion mechanisms.	Purpose, classification, source rights, retention policy, stewardship, and lawful instructions.
Models and agents	Gateway, runtime isolation, approved release controls, evaluation plumbing.	Allowed providers and packages, domain evaluation, human gates, budgets, and outcome review.

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41 Private Cloud Operating Envelope

Private Cloud places the customer intelligence domain in a dedicated customer or provider-controlled cloud boundary while preserving the same product contracts, artifact supply chain, and conformance evidence.

Isolation and integration

- Deploy into declared accounts, subscriptions, projects, networks, clusters, regions, and key boundaries owned according to contract.
- Integrate customer identity, keys, logging, security operations, private connectivity, registries, data services, and recovery controls through supported contracts.
- Permit management-plane connectivity only through declared, mutually authenticated, least-privilege endpoints; support relay, proxy, one-way, or offline promotion patterns where required.
- Keep operational telemetry minimized, classified, region-aware, and separable from customer intelligence content.
- Validate customer-specific infrastructure and security controls without forking canonical application semantics.

Operating model options

MODEL	ARCHITECTURE POSTURE
Vendor managed	The platform team operates within a dedicated boundary under customer-approved access, logging, key, and change controls.
Co-managed	Operational duties are divided by control domain; every duty has one accountable owner, interface, escalation, evidence, and continuity plan.
Customer managed	The customer operates approved artifacts and runbooks; support receives only explicitly authorized diagnostics and access.

PARITY RULE

A dedicated environment may add customer controls and integrations. It may not remove provenance, human authority, temporal truth, evaluation, audit, or portability obligations.

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42 On-Premise Operating Envelope

On-Premise operation supports customer-controlled facilities, restricted networks, sovereign environments, edge deployments, and fully disconnected enclaves without becoming a reduced-governance edition.

Disconnected architecture

The platform operates from locally available identity, policy, keys, evidence, models, packages, observability, documentation, licences, and time services. Dependencies that normally cross the boundary are mirrored through signed, inspectable bundles and controlled transfer procedures. Loss of external connectivity changes freshness and available capability, not truth labels or governance.

- Provide offline package, ontology, policy, model, connector, revocation, and vulnerability update bundles with chain of custody.
- Support local model resolution and deterministic fallbacks; declare when an approved cloud-dependent capability is unavailable.
- Maintain local audit continuity, usage accounting, evaluation, backups, recovery, and administrative evidence.
- Use store-and-forward collection and export with deduplication, ordering, custody, reconciliation, and conflict handling.
- Make time quality explicit where secure time synchronization is degraded or contested.

Environmental assumptions

CONSTRAINT	ARCHITECTURE RESPONSE
Air-gapped	No ambient external dependency; all required artifacts and authorities available locally or through governed transfer.
Intermittent link	Checkpoint, queue, deduplicate, reconcile, and expose external freshness without corrupting local continuity.
Limited capacity	Admission control, tiering, prioritized products, bounded agents, reduced models, and explicit service envelopes.
Customer hardware	Supported platform profile, preflight validation, capacity and accelerator discovery, health evidence, and conformance tests.
Restricted administration	Customer identity, keys, privileged access, break-glass, audit, and support bundle controls remain local.

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43 Sovereignty, Residency, Keys, and Disconnected Operation

Sovereignty is an enforceable set of boundaries over data, identities, keys, models, administration, telemetry, support, software supply, and external dependency—not a deployment label.

Sovereignty policy dimensions

DIMENSION	POLICY QUESTION
Data and metadata	Where may content, indexes, graph state, backups, logs, telemetry, and support artifacts be stored or processed?
Keys and secrets	Who owns, hosts, rotates, revokes, recovers, and audits data, signing, workload, and transport keys?
Models	Which providers, locations, weights, prompts, telemetry, retention, fine-tunes, and training uses are permitted?
Administration	Which identities, jurisdictions, networks, approvals, recording, and emergency procedures may administer the system?
Software supply	Who signs, verifies, mirrors, approves, scans, promotes, revokes, and rolls back artifacts and dependencies?
Support and observability	What operational data may leave the boundary, for what purpose, duration, receiver, and deletion obligation?
Continuity and exit	How are restore, portability, escrow, export, deletion, and independent operation proven?

Key separation

Data encryption, transport identity, artifact signing, audit integrity, backup protection, secret wrapping, and customer export use separated key roles. Key references are carried with protected objects and operations. Rotation and revocation are designed as normal lifecycle events with impact analysis and recovery, not exceptional maintenance.

> **SOVEREIGNTY RULE**
> Availability may fail closed when the only alternative would violate residency, key ownership, approved model, or administrative-boundary policy.

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44 Security Architecture

Security protects evidence integrity, strategic confidentiality, human authority, model and agent boundaries, operational continuity, and the ability to explain what occurred.

Security domains

DOMAIN	REQUIRED CONTROLS
Identity and privilege	Phishing-resistant assurance where warranted, workload identity, least privilege, continuous authorization, separation of duties, privileged access, and break-glass review.
Application and API	Schema validation, object authorization, rate and abuse controls, secure defaults, dependency isolation, secret handling, and security testing.
Data and knowledge	Classification, encryption, tenant isolation, object policy, minimization, integrity, retention, deletion, backup protection, and exfiltration controls.
AI and agents	Prompt-injection resistance, tool mediation, capability grants, context isolation, model evaluation, package sandboxing, budget, supervision, and output gating.
Supply chain	Provenance, signed artifacts, dependency inventory, build isolation, vulnerability management, promotion policy, revocation, and reproducible evidence.
Platform and network	Segmentation, hardened workload identity, egress control, runtime protection, patching, configuration policy, secure boot where applicable, and recovery.
Detection and response	Security telemetry, tamper-evident audit, threat analytics, investigation, containment, customer notification, forensics, and lessons governed into release.

Threat-informed architecture

- Assume external sources may contain malware, deception, poisoned data, hidden instructions, and adversarial narratives.
- Assume credentials, models, packages, connectors, users, administrators, and workloads may be compromised.
- Protect against cross-tenant access, unauthorized graph traversal, strategic inference leakage, model extraction, tool misuse, and covert exfiltration.
- Detect manipulation of evidence, time, provenance, forecasts, scenarios, simulations, recommendations, approvals, and audit records.
- Design containment at source, package, agent, model, service, tenant, cell, region, and deployment boundaries.

CONSTITUTIONAL INHERITANCE / C-035 · C-036 · C-037 · C-039 · C-040 · C-043 · C-050

45 Resilience and Safe Degradation

The platform preserves truthful operating state and human authority through partial failure. It may reduce freshness, speed, coverage, automation, or available methods; it must not invent confidence or silently bypass controls.

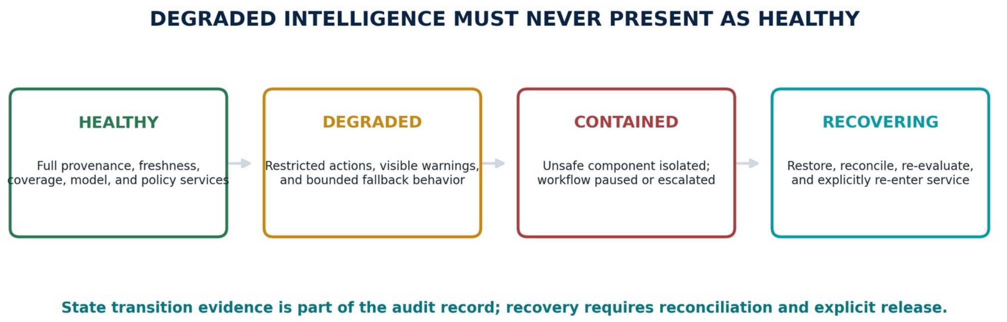

Figure 18. Operational state model for healthy, degraded, contained, and recovering intelligence services.

Degradation order

PRIORITY	PRESERVED CAPABILITY
1	Identity, tenant isolation, policy enforcement, evidence integrity, audit continuity, human authority, and safe stop.
2	Current warnings, active decisions, commitments, critical workflows, correction state, and operational health.
3	Evidence acquisition, graph and twin updates, priority forecasts, executive briefings, and customer-defined critical products.
4	Broader search, scenarios, simulations, agent automation, background evaluation, enrichment, and non-critical recomputation.
5	Performance optimizations, convenience features, and deferrable derived products.

Failure semantics

- Every user-visible product declares freshness, coverage, dependency health, partial state, last successful revision, and unavailable capability when material.
- Queues and workflows preserve causation, ordering requirements, idempotency, deadlines, and recovery position.
- The system fails closed for unauthorized access, unsafe external execution, missing final approval, or violated sovereignty policy.
- It may fail open only for explicitly classified non-sensitive availability functions under approved policy and with visible reduced assurance.
- Recovery reconciles canonical state, projections, indexes, external effects, and audit evidence before declaring full health.

RESILIENCE RULE
A stale answer clearly marked stale is a degraded product. A stale answer presented as current is an integrity failure.

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46 Observability and Site Reliability

Observability measures both platform mechanics and intelligence-product integrity. Service uptime alone cannot prove that strategic products are current, complete, explainable, or safe.

Operational signal model

SIGNAL CLASS	EXAMPLES
Service	Availability, latency, error, saturation, queue age, retry, dependency, capacity, and cost.
Data and evidence	Coverage, freshness, completeness, custody, quarantine, schema drift, correction lag, and orphan rate.
Intelligence product	Extraction quality, graph integrity, retrieval coverage, twin staleness, forecast calibration, scenario coherence, simulation reproducibility, decision completeness.
AI and agents	Model route, evaluation threshold, tool calls, policy denial, abstention, budget, anomaly, human override, and unsafe-output containment.
Security and governance	Identity assurance, privileged action, policy change, export, deletion, package promotion, separation-of-duties conflict, and audit continuity.
Experience	Time-to-understanding, attention precision, warning acknowledgement, workflow age, accessibility, explanation use, and user challenge.

SLO hierarchy

Platform SLOs govern technical services. Product SLOs govern the quality and timeliness of evidence, graph, twin, forecast, warning, decision, and briefing products. Mission or business SLOs govern customer-defined critical workflows and decision windows. Error budgets may reduce release velocity or non-critical computation; they may not waive security, sovereignty, decision authority, or truthfulness controls.

- Trace context follows a product from source contract through decision and outcome without placing sensitive content in global telemetry.
- Runbooks identify affected product semantics, not only failing infrastructure components.
- Synthetic and replay probes validate end-to-end evidence, authorization, correction, decision, export, and recovery paths.
- Operational incidents create governed impact records and Lesson Proposals when architecture behavior must change.

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47 Portability, Interoperability, and Exit

The Eye integrates into an enterprise without becoming an irreversible dependency. Customers retain authorized access to their data, knowledge, decisions, configuration, and provenance in documented, verifiable forms.

Interoperability surface

- Versioned APIs for commands, queries, explanations, administration, and export.
- Schema-registry-backed events and streams with compatibility metadata, ordering, replay, and correction semantics.
- Connector SDKs and source contracts that isolate external-system change from canonical objects.
- Ontology, graph, model, scenario, twin, agent, workflow, policy, and package manifests with signed provenance.
- Standards-aligned identity, cryptography, telemetry, geospatial, document, tabular, and structured export where semantics are sufficient.

Exit package

EXPORT DOMAIN	MINIMUM EXPORT
Evidence and memory	Authorized raw and derived objects, versions, metadata, classifications, retention state, checksums, and custody references.
Knowledge and strategy	Ontologies, entities, relationships, graph revisions, Strategy Graph, objectives, risks, opportunities, and provenance.
Twins and foresight	Twin definitions and snapshots, forecasts, scenarios, assumptions, indicators, simulation manifests, outputs, and validation.
Decisions and learning	Decision packages, approvals, dissent, commitments, outcomes, replays, lesson proposals, and release evidence.
Configuration	Policies where releasable, workflows, connector definitions, package manifests, evaluation specifications, and customer-owned models or prompts.
Verification	Inventory, schema versions, checksums, signatures, completeness report, unresolved references, deletion scope, and transfer audit.

Exit assurance

Export is tested as an operational capability, not improvised at termination. The customer can request a scoped export, validate completeness, import representative packages into an independent verifier, and obtain deletion evidence after contractual and legal obligations are satisfied. Proprietary implementation mechanisms may remain protected; customer-owned intelligence and the semantics required to interpret it do not become hostage to the platform.

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PART VI

Architecture Governance

Decision rights, conformity, quality contracts, evolution, specialization, verification, and documentation boundaries.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

48 Architecture Decision Rights

Architecture is governed through explicit authorities, recorded decisions, independent challenge, conformance evidence, and constitutional escalation. No team can silently trade away a frozen property for local delivery convenience.

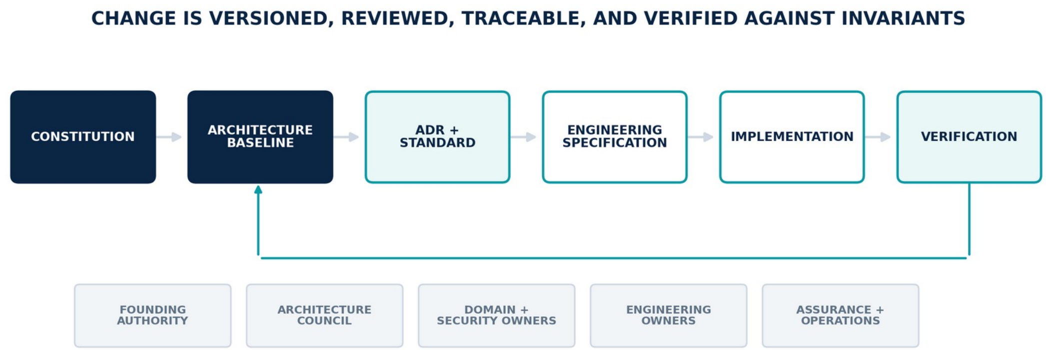

Figure 19. Architecture governance chain from constitution through implementation and verification.

Decision authorities

AUTHORITY	ACCOUNTABILITY
Product Constitution Council	Ratifies changes to constitutional invariants, category, human authority, canonical layers, deployment parity, and responsible-use boundaries.
Architecture Council	Owns Volume 3, cross-domain contracts, canonical objects, quality attributes, trust boundaries, ADR standards, and conformance decisions.
Domain Architecture Owners	Own layer and control-plane designs within the canonical boundary and maintain component, interface, event, data, security, and failure semantics.
Platform and SRE Authority	Owns deployability, capacity, resilience, observability, recovery, operational security, and service acceptance across deployment modes.
AI Governance Authority	Owns model, agent, evaluation, package, safety, and learning release controls without receiving strategic decision rights.
Data and Ontology Governance	Owns source, information quality, identity, ontology, graph, retention, correction, privacy, and lifecycle stewardship.
Customer Governance Authority	Sets customer purpose, policy, sovereignty, role, data, risk, approval, marketplace, and use boundaries within the product constitution.

Decision classes

- Constitutional: changes a C-001–C-052 commitment and requires formal ratification plus downstream document revision.
- Cross-domain: changes a canonical object, interface, truth or time semantic, provenance, policy, human gate, or deployment parity and requires Architecture Council approval.
- Domain: changes an implementation within an approved contract and is owned by the relevant architecture authority.
- Operational: changes configuration within an approved runbook, policy, capacity, and risk envelope and remains auditable.
- Emergency: contains an active incident under pre-authorized scope and must receive retrospective review before it becomes normal design.

CONSTITUTIONAL INHERITANCE / C-035 · C-036 · C-051 · C-052

49 Conformance and Exceptions

Conformance is demonstrated by evidence across structure, behavior, security, quality, operations, and user-visible semantics. An exception is a governed, temporary risk decision—not a parallel architecture.

Conformance evidence

EVIDENCE FAMILY	REQUIRED PROOF
Contract	Schema, API, event, workflow, object, ontology, policy, error, and compatibility tests.
Trust	Identity, authorization, provenance, temporal truth, correction, audit, privacy, classification, and deletion tests.
Human authority	Approval, delegation, dissent, override, separation of duties, external execution, and replay tests.
Quality	SLOs, evaluation thresholds, degradation markers, calibration, reproducibility, explanation, and accessibility evidence.
Deployment	SaaS, Private Cloud, and On-Premise semantic fixtures plus disconnected and sovereignty-boundary tests.
Operations	Capacity, failover, restore, reconciliation, incident, rollback, observability, and exit exercises.

Exception record

An exception names the unmet requirement, system and customer scope, reason, evidence, risk, affected invariants, compensating controls, owner, approvers, start, expiry, monitoring, exit criteria, and remediation plan. Constitutional boundaries cannot be waived through ordinary exception. Expiry stops the affected promotion or operation unless the authorized body renews the exception with current evidence.

GOVERNANCE RULE

Repeated exceptions against the same contract are architecture evidence. They trigger remediation or formal change review; they do not quietly redefine the baseline.

CONSTITUTIONAL INHERITANCE / C-035 · C-036 · C-042 · C-049 · C-051

50 Quality Attribute Contracts

Quality is expressed as measurable behavior for intelligence products and operating services. Targets vary by customer, domain, and criticality; definitions and failure semantics remain common.

Architecture quality model

ATTRIBUTE	ARCHITECTURE DEFINITION
Correctness	Outputs preserve schema, semantic, temporal, provenance, policy, numerical, and workflow invariants.
Trustworthiness	Evidence, uncertainty, explanation, accountability, contestability, security, privacy, and audit are complete for intended use.
Freshness and coverage	Products declare source field of view, observation recency, update latency, missing regions or domains, and downstream staleness.
Reliability	The service and product meet availability, durability, ordering, recovery, idempotency, and safe-degradation commitments.
Performance	End-to-end latency, throughput, concurrency, query complexity, agent budget, and decision window meet declared classes.
Scalability	Cells, partitions, graph traversals, streams, simulations, and models scale without changing isolation or meaning.
Portability	Artifacts, data, knowledge, configuration, and conformance fixtures operate across the three deployment envelopes.
Usability and accessibility	Roles can understand state, evidence, uncertainty, control, and required action through supported access modes.
Operability	Health, impact, capacity, cost, change, failure, recovery, and support boundaries are visible and actionable.

Quality contract structure

- Product or service name, owner, consumers, criticality, deployment scope, and business or mission purpose.
- Indicator definition, units, population, segmentation, measurement point, exclusions, and source of truth.
- Target, window, budget, warning, breach, escalation, review, and customer-visible communication.
- Dependencies, assumptions, partial-state behavior, safe degradation, recovery objective, and validation method.
- Release gate, operational dashboard, audit evidence, exception process, and improvement owner.

CONSTITUTIONAL INHERITANCE / C-005 · C-014 · C-034 · C-043 · C-045 · C-047 · C-048 · C-049

51 Release, Configuration, and Schema Evolution

Every material behavior of The Eye is a versioned, signed, evaluable release artifact with compatibility, provenance, promotion, monitoring, rollback, and retirement semantics.

Artifact classes

ARTIFACT	REQUIRED LIFECYCLE CONTROLS
Application and infrastructure	Source and build provenance, dependencies, security evidence, configuration schema, migration, conformance, signed promotion, rollback, and support window.
Canonical schema and events	Semantic version, compatibility class, fixtures, consumer inventory, migration, dual operation where required, deprecation, and archival decoder.
Ontology and graph rules	Proposal, steward review, affected identities and queries, migration, reconciliation, domain compatibility, validation, and reversible revision.
Model, prompt, retrieval, and agent	Task contract, data boundary, evaluation, safety, package manifest, model card, rollout, runtime monitoring, quarantine, and rollback.
Policy and workflow	Authority, human impact, simulation, separation of duties, in-flight instance behavior, approval, effective time, audit, and emergency reversal.
Twin, scenario, and simulation package	Dependencies, assumptions, version resolver, validation, customer localization, compatibility, retirement, and replay preservation.

Promotion chain

Artifacts move through development, verification, security and policy review, representative evaluation, integration, deployment-mode conformance, pre-production, controlled rollout, production, monitoring, and retirement. Promotion is evidence-based and environment-specific. The exact artifact and intent remain unchanged; environment binding supplies approved identities, secrets, capacity, regions, endpoints, and policy.

- Breaking changes require an explicit migration and consumer cutover plan.
- In-flight decisions, workflows, simulations, and agent executions retain their resolved versions unless deliberately migrated.
- Rollback includes schemas, state, projections, packages, models, policy, and audit continuity—not just binaries.
- Emergency changes are bounded and reviewed before they enter the supported baseline.
- End-of-life preserves the ability to decode, explain, replay, export, and audit historical artifacts.

CONSTITUTIONAL INHERITANCE / C-011 · C-025 · C-031 · C-035 · C-038 · C-045

52 Tenant and Domain Specialization

The Eye serves governments, defence, banks, healthcare, manufacturing, energy, logistics, telecom, retail, consulting, and SMEs through governed specialization over one universal core.

Permitted extension surfaces

SURFACE	EXTENSION CONTRACT
Sources and connectors	Customer-authorized adapters, source contracts, schemas, mappings, quality, rights, and correction behavior.
Ontology and graph	Domain types, relationships, constraints, stewardship, compatibility, provenance, and cross-domain mapping.
Digital Twins	Domain twin definitions, state variables, topology, estimators, constraints, models, measures, and validation.
Forecast, scenario, simulation	Targets, drivers, assumptions, indicators, domain models, solvers, interventions, evaluation, and packages.
Decision and workflow	Decision templates, criteria, obligations, approvals, playbooks, commitments, review cadence, and outcomes.
Agents and models	Domain agent packages, prompts, tools, models, policies, evaluations, budgets, human gates, and rollback.
Experience	Role workspaces, vocabulary, briefing templates, accessibility modes, views, and notification policy.

Non-permitted specialization

- Removing final human authority or allowing an agent to become the accountable decision-maker.
- Changing canonical truth, temporal, provenance, identity, correction, audit, or Decision Replay semantics.
- Creating a deployment mode with different meaning for the ten layers or canonical objects.
- Creating a parallel tenant-specific object model that cannot participate in shared governance, export, explanation, and conformance.
- Hiding uncertainty, dissent, source health, missing evidence, or degraded capability to satisfy a domain presentation preference.

EXTENSION RULE

A domain may add semantics and controls. It may not subtract the evidence needed for trust, human authority, temporal reconstruction, or enterprise portability.

CONSTITUTIONAL INHERITANCE / C-004 · C-007 · C-038 · C-042 · C-047 · C-048

53 Architecture Verification

The architecture is considered real only when representative end-to-end journeys, adversarial conditions, deployment modes, recovery paths, and exit procedures can demonstrate its contracts with machine-verifiable evidence.

Reference verification journeys

JOURNEY	MUST PROVE
Observe to warning	Authorized acquisition, custody, transformation, graph update, weak-signal detection, consequence, attention policy, explanation, and closure.
Evidence to decision	Truth-state separation, temporal context, provenance, forecast, scenarios, simulation, alternatives, dissent, human approval, commitment, and replay.
Correction	Versioned correction, forward impact, invalidation, recomputation, owner review, historical preservation, and propagation evidence.
Agent workflow	Identity, purpose, planning, capability grant, tool control, evidence, evaluation, supervision, interruption, human gate, and audit.
Isolation and sovereignty	Cross-tenant denial, regional and key constraints, admin boundary, minimized telemetry, provider restriction, and export policy.
Failure and recovery	Partial dependency loss, truthful degradation, queue continuity, safe stop, restore, reconciliation, and product health.
Deployment parity	Equivalent object, workflow, decision, provenance, correction, package, and export semantics in all three modes.
Exit	Complete scoped export, independent validation, unresolved-reference report, authorized deletion, and audit evidence.

Verification gates

- Design review demonstrates traceability to constitutional invariants and quality attributes.
- Contract verification proves schemas, APIs, events, objects, policies, workflows, and compatibility.
- Security and responsible-use verification challenge identity, isolation, agents, models, data, supply chain, approvals, and external side effects.
- Performance and resilience verification exercises scale, load, dependency failure, regional or site loss, recovery, and backlog reconciliation.
- Customer acceptance validates role, domain, sovereignty, integration, accessibility, support, and operational responsibility boundaries.
- Continuous conformance monitors production artifacts, configuration, policy, drift, SLOs, exceptions, and evidence expiry.

CONSTITUTIONAL INHERITANCE / C-004 · C-005 · C-035 · C-042 · C-043 · C-045 · C-049 · C-050

54 Boundaries to Volumes 4–7

Volume 3 freezes the logical and operational contracts. Volumes 4–7 convert them into buildable specifications while preserving one vocabulary, one object system, one trust model, and one strategic loop.

Documentation inheritance

VOLUME	DERIVED RESPONSIBILITY	MAY NOT REDEFINE
Volume 4 — Engineering Specification	Service boundaries, APIs, events, workflows, algorithms, state machines, error semantics, test plans, and delivery standards.	Layer responsibility, canonical object meaning, human gates, truth, time, provenance, deployment parity.
Volume 5 — AI Architecture	Model and agent runtime, routing, prompts, retrieval, tools, memory, evaluation, safety, supervision, and learning implementation.	Bounded authority, model pluralism, no silent self-modification, explainability, human final decision.
Volume 6 — Infrastructure Architecture	Physical topology, compute, network, storage, accelerators, platform services, deployment automation, SRE, security, continuity, and capacity.	Customer domain boundary, sovereignty, semantic parity, safe degradation, portability, audit continuity.
Volume 7 — Data Platform	Storage engines, schemas, streaming, lake and warehouse patterns, graph, indexing, lineage, quality, retention, migration, and data operations.	Canonical objects, temporal truth, truth-state separation, provenance, customer control, correction, export.

Required derivation package

- Every specification cites its controlling Volume 3 chapter, component ID, object ID, interface or event ID, quality contract, trust boundary, and constitutional invariant.
- Every implementation choice states its failure, degradation, recovery, observability, security, privacy, sovereignty, and portability behavior.
- Every AI component states its owner, purpose, authority, data boundary, model and prompt versions, tool permissions, evaluation, human gates, and rollback.
- Every storage and interface design preserves event, observation, valid, and system time plus exact provenance and policy revision.
- Every deployment design proves SaaS, Private Cloud, and On-Premise semantic parity or declares an approved capability limitation without changing meaning.

BASELINE STATUS

This volume is the controlled architecture baseline. Downstream detail is valid only while it conforms to this volume and the Product Constitution. A conflict requires correction or formal change control; it is not an implementation preference.

CONSTITUTIONAL INHERITANCE / C-001 · C-006 · C-008 · C-051 · C-052

PART APP

Reference Appendices

Normative component, object, interface, relationship, traceability, terminology, and architecture-decision references.

THE EYE / THE OPERATING SYSTEM FOR STRATEGIC INTELLIGENCE

A Logical Component Catalog

Component identifiers are stable architecture references. Volume 4 may decompose a logical component into services or combine implementation units, provided ownership, contracts, state, controls, and failure semantics remain intact.

ID	LAYER	LOGICAL COMPONENT	ARCHITECTURE RESPONSIBILITY
L1-C01	World Observation Layer	Source Registry and Contract Service	Owns versioned source registry and contract definitions, resolution, compatibility, stewardship state, and lifecycle events within World Observation Layer.
L1-C02	World Observation Layer	Connector Runtime and Adapter SDK	Mediates connector runtime and adapter sdk through authenticated, versioned, policy-evaluated contracts without leaking external semantics into the layer core.
L1-C03	World Observation Layer	Collection Scheduler and Subscription Manager	Coordinates collection scheduler and subscription lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L1-C04	World Observation Layer	Streaming and Batch Receivers	Owns the streaming and batch receivers capability inside World Observation Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L1-C05	World Observation Layer	Raw Preservation and Evidence Vault	Persists authoritative raw preservation and evidence vault state with version, integrity, policy, retention, and recovery semantics.
L1-C06	World Observation Layer	Authenticity and Chain-of-Custody Service	Owns the authenticity and chain-of-custody service capability inside World Observation Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L1-C07	World Observation Layer	Quarantine and Malware Inspection	Owns the quarantine and malware inspection capability inside World Observation Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L1-C08	World Observation Layer	Source Health, Freshness, and Coverage Monitor	Measures source health, freshness, and coverage monitor health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.
L1-C09	World Observation Layer	Correction and Withdrawal Intake	Owns the correction and withdrawal intake capability inside World Observation Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C01	Intelligence Layer	Format Detection, OCR, Transcription, and Parsing	Owns the format detection, ocr, transcription, and parsing capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C02	Intelligence Layer	Cleaning, Normalization, and Language Services	Owns the cleaning, normalization, and language services capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C03	Intelligence Layer	Classification and Routing	Owns the classification and routing capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C04	Intelligence Layer	Summarization and Structured Extraction	Owns the summarization and structured extraction capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C05	Intelligence Layer	Named-Entity Recognition and Candidate Linking	Owns the named-entity recognition and candidate linking capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.

ID	LAYER	LOGICAL COMPONENT	ARCHITECTURE RESPONSIBILITY
L2-C06	Intelligence Layer	Relationship, Event, and Claim Extraction	Owns the relationship, event, and claim extraction capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C07	Intelligence Layer	Contradiction and Corroboration Detection	Owns the contradiction and corroboration detection capability inside Intelligence Layer, including contract validation, versioned behavior, operational evidence, and failure state.
L2-C08	Intelligence Layer	Confidence and Truth-State Assessor	Executes the governed confidence and truth-state assessor method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L2-C09	Intelligence Layer	Human Review Queue and Adjudication	Coordinates durable human review queue and adjudication transitions, human tasks, deadlines, separation of duties, challenge, and audit evidence.
L2-C10	Intelligence Layer	Transformation Registry and Evaluation Harness	Owns versioned transformation registry and evaluation harness definitions, resolution, compatibility, stewardship state, and lifecycle events within Intelligence Layer.
L3-C01	Enterprise Memory	Memory Object Service	Owns the memory object service capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C02	Enterprise Memory	Document and Artifact Repository	Persists authoritative document and artifact repository state with version, integrity, policy, retention, and recovery semantics.
L3-C03	Enterprise Memory	Temporal Record Store	Persists authoritative temporal record store state with version, integrity, policy, retention, and recovery semantics.
L3-C04	Enterprise Memory	Version and Correction Ledger	Owns the version and correction ledger capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C05	Enterprise Memory	Retention, Legal Hold, and Deletion Service	Owns the retention, legal hold, and deletion service capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C06	Enterprise Memory	Permission-Aware Semantic Index	Owns the permission-aware semantic index capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C07	Enterprise Memory	Memory Retrieval and Context Assembly	Owns the memory retrieval and context assembly capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L3-C08	Enterprise Memory	Archive and Cold-Tier Manager	Coordinates archive and cold-tier lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L3-C09	Enterprise Memory	Memory Integrity and Orphan Detection	Owns the memory integrity and orphan detection capability inside Enterprise Memory, including contract validation, versioned behavior, operational evidence, and failure state.
L4-C01	Knowledge Graph	Ontology Registry and Governance Workflow	Owns versioned ontology registry and governance workflow definitions, resolution, compatibility, stewardship state, and lifecycle events within Knowledge Graph.
L4-C02	Knowledge Graph	Identity Resolution and Entity Stewardship	Owns the identity resolution and entity stewardship capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.
L4-C03	Knowledge Graph	Graph Write Validator	Owns the graph write validator capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.

ID	LAYER	LOGICAL COMPONENT	ARCHITECTURE RESPONSIBILITY
L4-C04	Knowledge Graph	Temporal Graph Store	Persists authoritative temporal graph store state with version, integrity, policy, retention, and recovery semantics.
L4-C05	Knowledge Graph	Provenance and Evidence Edge Service	Owns the provenance and evidence edge service capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.
L4-C06	Knowledge Graph	Graph Query and Traversal Service	Owns the graph query and traversal service capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.
L4-C07	Knowledge Graph	Reasoning and Constraint Engine	Executes the governed reasoning and constraint method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L4-C08	Knowledge Graph	Graph Change Subscription Service	Owns the graph change subscription service capability inside Knowledge Graph, including contract validation, versioned behavior, operational evidence, and failure state.
L4-C09	Knowledge Graph	Merge, Split, and Reconciliation Manager	Coordinates merge, split, and reconciliation lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L4-C10	Knowledge Graph	Graph Quality and Orphan Monitor	Measures graph quality and orphan monitor health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.
L5-C01	Digital Twins	Twin Registry and Boundary Service	Owns versioned twin registry and boundary definitions, resolution, compatibility, stewardship state, and lifecycle events within Digital Twins.
L5-C02	Digital Twins	State Estimation and Reconciliation	Owns the state estimation and reconciliation capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.
L5-C03	Digital Twins	Model Binding and Operating-Envelope Registry	Owns versioned model binding and operating-envelope registry definitions, resolution, compatibility, stewardship state, and lifecycle events within Digital Twins.
L5-C04	Digital Twins	Snapshot and Version Manager	Coordinates snapshot and version lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L5-C05	Digital Twins	Branch and Time-Travel Manager	Coordinates branch and time-travel lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L5-C06	Digital Twins	Twin Dependency Mapper	Owns the twin dependency mapper capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.
L5-C07	Digital Twins	Behavior and Constraint Runtime	Owns the behavior and constraint runtime capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.
L5-C08	Digital Twins	Validation and Calibration Service	Owns the validation and calibration service capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.
L5-C09	Digital Twins	Synthetic-State Labelling and Isolation	Owns the synthetic-state labelling and isolation capability inside Digital Twins, including contract validation, versioned behavior, operational evidence, and failure state.
L6-C01	Prediction Engine	Forecast Registry and Horizon Service	Owns versioned forecast registry and horizon definitions, resolution, compatibility, stewardship state, and lifecycle events within Prediction Engine.
L6-C02	Prediction Engine	Feature and Context Assembler	Owns the feature and context assembler capability inside Prediction Engine, including contract validation, versioned behavior, operational evidence, and failure state.

ID	LAYER	LOGICAL COMPONENT	ARCHITECTURE RESPONSIBILITY
L6-C03	Prediction Engine	Model Gateway and Candidate Router	Mediates model gateway and candidate router through authenticated, versioned, policy-evaluated contracts without leaking external semantics into the layer core.
L6-C04	Prediction Engine	Ensemble and Disagreement Manager	Coordinates ensemble and disagreement lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L6-C05	Prediction Engine	Calibration and Uncertainty Service	Owns the calibration and uncertainty service capability inside Prediction Engine, including contract validation, versioned behavior, operational evidence, and failure state.
L6-C06	Prediction Engine	Driver Attribution and Sensitivity Engine	Executes the governed driver attribution and sensitivity method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L6-C07	Prediction Engine	Forecast Evaluation and Backtesting	Measures forecast evaluation and backtesting health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.
L6-C08	Prediction Engine	Drift and Refresh Monitor	Measures drift and refresh monitor health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.
L6-C09	Prediction Engine	Forecast Version Store	Persists authoritative forecast version store state with version, integrity, policy, retention, and recovery semantics.
L7-C01	Scenario Intelligence	Scenario Registry and Portfolio Manager	Owns versioned scenario registry and portfolio definitions, resolution, compatibility, stewardship state, and lifecycle events within Scenario Intelligence.
L7-C02	Scenario Intelligence	Assumption and Hypothesis Register	Owns versioned assumption and hypothesis register definitions, resolution, compatibility, stewardship state, and lifecycle events within Scenario Intelligence.
L7-C03	Scenario Intelligence	Branch and Path Engine	Executes the governed branch and path method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L7-C04	Scenario Intelligence	Actor, Driver, and Mechanism Model	Owns the actor, driver, and mechanism model capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L7-C05	Scenario Intelligence	Indicator and Signpost Service	Owns the indicator and signpost service capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L7-C06	Scenario Intelligence	Scenario Coherence Validator	Owns the scenario coherence validator capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L7-C07	Scenario Intelligence	Intervention and Impact Mapper	Owns the intervention and impact mapper capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L7-C08	Scenario Intelligence	Scenario Comparison and Review Workspace	Coordinates durable scenario comparison and review workspace transitions, human tasks, deadlines, separation of duties, challenge, and audit evidence.
L7-C09	Scenario Intelligence	Scenario Package Import and Localization	Owns the scenario package import and localization capability inside Scenario Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L8-C01	Simulation Engine	Experiment Registry and Run Planner	Owns versioned experiment registry and run planner definitions, resolution, compatibility, stewardship state, and lifecycle events within Simulation Engine.
L8-C02	Simulation Engine	Twin, Scenario, and Model Version Resolver	Executes the governed twin, scenario, and model version resolver method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.

ID	LAYER	LOGICAL COMPONENT	ARCHITECTURE RESPONSIBILITY
L8-C03	Simulation Engine	Constraint and Intervention Engine	Executes the governed constraint and intervention method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L8-C04	Simulation Engine	Seed and Determinism Manager	Coordinates seed and determinism lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L8-C05	Simulation Engine	Solver and Simulator Adapter Fabric	Mediates solver and simulator adapter fabric through authenticated, versioned, policy-evaluated contracts without leaking external semantics into the layer core.
L8-C06	Simulation Engine	Distributed Run Orchestrator	Coordinates distributed run orchestrator lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L8-C07	Simulation Engine	Impact and Second-Order Effect Analyzer	Executes the governed impact and second-order effect analyzer method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L8-C08	Simulation Engine	Sensitivity and Robustness Service	Owns the sensitivity and robustness service capability inside Simulation Engine, including contract validation, versioned behavior, operational evidence, and failure state.
L8-C09	Simulation Engine	Reproducibility and Validation Store	Persists authoritative reproducibility and validation store state with version, integrity, policy, retention, and recovery semantics.
L9-C01	Decision Intelligence	Decision Registry and Context Resolver	Owns versioned decision registry and context resolver definitions, resolution, compatibility, stewardship state, and lifecycle events within Decision Intelligence.
L9-C02	Decision Intelligence	Objective, Constraint, and Obligation Service	Owns the objective, constraint, and obligation service capability inside Decision Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L9-C03	Decision Intelligence	Option and Alternative Manager	Coordinates option and alternative lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L9-C04	Decision Intelligence	Evidence and Foresight Assembler	Owns the evidence and foresight assembler capability inside Decision Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L9-C05	Decision Intelligence	Criteria, Trade-off, and Information-Value Engine	Executes the governed criteria, trade-off, and information-value method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.
L9-C06	Decision Intelligence	Recommendation and Explanation Service	Owns the recommendation and explanation service capability inside Decision Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L9-C07	Decision Intelligence	Approval, Dissent, and Override Workflow	Coordinates durable approval, dissent, and override workflow transitions, human tasks, deadlines, separation of duties, challenge, and audit evidence.
L9-C08	Decision Intelligence	Decision Record and Commitment Service	Owns the decision record and commitment service capability inside Decision Intelligence, including contract validation, versioned behavior, operational evidence, and failure state.
L9-C09	Decision Intelligence	Monitoring Contract and Review Scheduler	Measures monitoring contract and review scheduler health and fitness, emits decision-active quality state, and initiates exception or remediation workflow.
L10-C01	Executive Operating System	Executive Workspace and Responsibility Model	Presents policy-filtered executive workspace and responsibility model products with evidence, uncertainty, authority, state, and required action intact.
L10-C02	Executive Operating System	Attention Policy and Materiality Engine	Executes the governed attention policy and materiality method against declared inputs and versions; returns results, uncertainty, evaluation, and lineage.

ID	LAYER	LOGICAL COMPONENT	ARCHITECTURE RESPONSIBILITY
L10-C03	Executive Operating System	Live Briefing Service	Presents policy-filtered live briefing products with evidence, uncertainty, authority, state, and required action intact.
L10-C04	Executive Operating System	Decision and Scenario Rooms	Presents policy-filtered decision and scenario rooms products with evidence, uncertainty, authority, state, and required action intact.
L10-C05	Executive Operating System	Objective, Commitment, and Outcome Review	Coordinates durable objective, commitment, and outcome review transitions, human tasks, deadlines, separation of duties, challenge, and audit evidence.
L10-C06	Executive Operating System	Strategic Cadence Scheduler	Coordinates strategic cadence scheduler lifecycle, admission, ordering, retries, budgets, escalation, completion, and observable state.
L10-C07	Executive Operating System	Notification and Escalation Service	Owns the notification and escalation service capability inside Executive Operating System, including contract validation, versioned behavior, operational evidence, and failure state.
L10-C08	Executive Operating System	Role-Appropriate Access Modes	Owns the role-appropriate access modes capability inside Executive Operating System, including contract validation, versioned behavior, operational evidence, and failure state.
L10-C09	Executive Operating System	Executive Search and Explanation	Owns the executive search and explanation capability inside Executive Operating System, including contract validation, versioned behavior, operational evidence, and failure state.

B Canonical Intelligence Object Catalog

Every object inherits the universal header defined in Chapter 7. The owning layer controls authoritative lifecycle; other layers consume versioned references and policy-filtered projections.

CODE	OBJECT	OWNING LAYER	DEFINITION
SRC	Source Contract	Observation	Authority, rights, purpose, schema, latency, quality, authenticity, correction behavior, and ownership for a source.
OBS	Observation Record	Observation	An immutable record that a source produced content or state at a specific observation time.
EVD	Evidence Object	Observation / Intelligence	Preserved source material plus chain of custody, transformations, classification, quality, and coverage context.
CLM	Claim	Intelligence	A source-attributed assertion with truth state, confidence, temporal scope, and supporting or contradicting evidence.
EVT	Event	Intelligence	A temporally scoped occurrence connected to actors, locations, assets, mechanisms, and evidence.
ENT	Entity	Knowledge Graph	A governed identity representing a company, country, person, product, technology, market, asset, objective, or other canonical thing.
REL	Relationship	Knowledge Graph	A typed, temporal, provenance-bearing connection between governed entities or intelligence objects.
ASM	Assessment	Intelligence	A versioned analytical judgment that separates evidence, method, uncertainty, dissent, and accountable author.
MEM	Memory Entry	Enterprise Memory	A durable, permission-aware, versioned organizational record with retention and correction semantics.
GRV	Graph Revision	Knowledge Graph	An atomic, validated set of node, edge, ontology, provenance, and temporal changes.
OBJ	Objective	Strategy Graph	A governed strategic intent with owner, horizon, measures, constraints, dependencies, and review cadence.
RSK	Risk / Opportunity	Strategy Graph	A symmetric strategic exposure connected to drivers, affected objectives, scenarios, responses, and owners.
TWN	Twin Snapshot	Digital Twins	A versioned state of a declared system boundary, grounded in evidence, graph identity, models, assumptions, and validity.
FCT	Forecast Package	Prediction	A horizon-specific probability distribution with drivers, calibration, sensitivity, model lineage, and failure conditions.
SCN	Scenario Branch	Scenario	A coherent future branch with baseline, assumptions, actors, drivers, indicators, interventions, impacts, and review state.
SIM	Simulation Run	Simulation	A reproducible experiment with initial state, model and twin versions, constraints, seeds, interventions, outputs, and sensitivity.
DEC	Decision Package	Decision	A complete choice environment containing objectives, options, evidence, forecasts, scenarios, simulations, uncertainty, dissent, approvals, and monitoring.
CMT	Commitment	Decision / Executive	An approved institutional promise with owner, resources, obligations, execution state, indicators, review dates, and closure criteria.
OUT	Outcome Record	Executive / Learning	Observed consequences of an approved decision, linked to actions, context, later evidence, and evaluation.
LES	Lesson Proposal	Learning	A proposed, evidence-backed change to institutional memory, model, policy, workflow, ontology, or practice pending governance.
WRN	Warning Product	Proactive Intelligence	A deduplicated, prioritized alert with evidence, confidence, consequence, response window, owner, and closure criteria.
AGX	Agent Execution	Agent System	A bounded execution trace with agent identity, owner, purpose, plan, tools, data scope, budgets, outputs, evaluation, and stop state.

CODE	OBJECT	OWNING LAYER	DEFINITION
POL	Policy Decision	Control Plane	A recorded authorization, obligation, denial, exception, or purpose decision bound to identity, object, action, and time.
AUD	Audit Evidence	Control Plane	Tamper-evident proof of material access, transformation, model or agent action, approval, decision, release, correction, or export.

Universal object obligations

- Stable object and tenant identity; explicit schema and object versions; immutable prior history.
- Event, observation, valid, and system time where meaningful; current and historical projections remain distinguishable.
- Truth state, confidence, uncertainty, synthetic-state marker, and lifecycle status.
- Evidence, source, transformation, model, agent, human, policy, causation, and supersession references.
- Classification, purpose, owner, access policy, retention, quality, freshness, and expiry.
- Correlation, causation, trace, idempotency, audit, correction, export, and deletion evidence.

C Interface and Event Catalog

This catalog defines the canonical layer-facing interaction surface. Volume 4 assigns payload schemas, protocols, reliability classes, error models, and service ownership without changing the semantics below.

ID	LAYER	INTERFACE	STYLE	CONTRACT
L1-I01	World Observation Layer	RegisterSource	Command	Creates or versions a governed source contract after authorization and rights checks.
L1-I02	World Observation Layer	Acquire	Command / stream	Collects content or state under contract, purpose, rate, and residency constraints.
L1-I03	World Observation Layer	ObservationRecorded	Domain event	Announces an immutable observation and evidence reference.
L1-I04	World Observation Layer	SourceHealthChanged	Domain event	Reports freshness, completeness, authenticity, latency, or coverage state changes.
L1-I05	World Observation Layer	CorrectionReceived	Domain event	Introduces source correction, withdrawal, or supersession without destroying prior history.
L2-I01	Intelligence Layer	TransformEvidence	Command	Runs a declared transformation plan against immutable evidence.
L2-I02	Intelligence Layer	IntelligenceObject Proposed	Domain event	Publishes a claim, event, entity, relationship, or assessment candidate with lineage.
L2-I03	Intelligence Layer	ContradictionDetected	Domain event	Links incompatible assertions without collapsing them into a single truth state.
L2-I04	Intelligence Layer	ReviewRequested	Workflow event	Routes ambiguous, high-impact, low-confidence, or policy-sensitive output to accountable humans.
L2-I05	Intelligence Layer	TransformationEvaluated	Domain event	Records quality, safety, cost, latency, and fitness for the producing version.
L3-I01	Enterprise Memory	CommitMemory	Command	Persists a governed object version with policy, provenance, classification, and temporal metadata.
L3-I02	Enterprise Memory	RetrieveContext	Query	Returns policy-filtered memory and explanation links for an explicit purpose.
L3-I03	Enterprise Memory	MemoryCorrected	Domain event	Versions a correction and identifies affected downstream objects.
L3-I04	Enterprise Memory	RetentionActionDue	Domain event	Initiates archive, review, legal hold, deletion, or customer export workflow.
L3-I05	Enterprise Memory	DeletionVerified	Audit event	Proves scope, authorization, execution, and residual references for a deletion action.
L4-I01	Knowledge Graph	ResolveIdentity	Command / query	Returns or proposes canonical identity with evidence, confidence, and stewardship state.
L4-I02	Knowledge Graph	CommitGraphRevision	Command	Applies an atomic, validated node, edge, ontology, and provenance change set.
L4-I03	Knowledge Graph	GraphChanged	Domain event	Publishes affected identities, relationships, temporal scopes, and downstream subscriptions.
L4-I04	Knowledge Graph	TraverseContext	Query	Returns policy-filtered paths with provenance, time, truth state, and confidence.
L4-I05	Knowledge Graph	OntologyChange Proposed	Workflow event	Initiates compatibility, migration, domain, and governance review.
L5-I01	Digital Twins	CreateTwinDefinition	Command	Declares boundary, identity, evidence, behaviors, owners, validation, and limitations.
L5-I02	Digital Twins	ReconcileTwin	Command	Produces a new observed-state snapshot from evidence and graph revisions.
L5-I03	Digital Twins	BranchTwin	Command	Creates an isolated synthetic branch for scenario or simulation use.
L5-I04	Digital Twins	TwinStateChanged	Domain event	Publishes version, changed variables, confidence, freshness, and dependency impacts.

ID	LAYER	INTERFACE	STYLE	CONTRACT
L5-I05	Digital Twins	ValidateTwin	Command / event	Evaluates state and model fitness inside the declared operating envelope.
L6-I01	Prediction Engine	RequestForecast	Command	Creates or refreshes a forecast under a declared objective, target, horizon, context, and policy.
L6-I02	Prediction Engine	ForecastProduced	Domain event	Publishes distribution, confidence, calibration, drivers, lineage, and expiry.
L6-I03	Prediction Engine	ForecastFitnessChanged	Domain event	Signals drift, calibration failure, data shift, or operating-envelope breach.
L6-I04	Prediction Engine	BacktestForecast	Command	Evaluates historical versions without contaminating prior decision context.
L6-I05	Prediction Engine	ForecastWithdrawn	Domain event	Marks a version unfit and propagates impact to scenarios, warnings, simulations, and decisions.
L7-I01	Scenario Intelligence	CreateScenario	Command	Declares baseline, branch logic, assumptions, drivers, owners, indicators, and review policy.
L7-I02	Scenario Intelligence	BranchScenario	Command	Creates a versioned upside, downside, disruption, or user-defined alternative.
L7-I03	Scenario Intelligence	ScenarioIndicator Changed	Domain event	Updates branch relevance when a monitored signpost moves.
L7-I04	Scenario Intelligence	ScenarioCoherence Failed	Domain event	Identifies internal inconsistency, invalid assumption, or incompatible dependency.
L7-I05	Scenario Intelligence	ScenarioReviewed	Workflow event	Records human review, dissent, continuation, retirement, or promotion to simulation.
L8-I01	Simulation Engine	SubmitExperiment	Command	Validates declared state, versions, constraints, budgets, policy, and expected outputs.
L8-I02	Simulation Engine	SimulationStarted	Domain event	Records resolved artifacts, execution identity, environment, seed, and run state.
L8-I03	Simulation Engine	SimulationCompleted	Domain event	Publishes outputs, impacts, uncertainty, sensitivity, validation, and resource evidence.
L8-I04	Simulation Engine	ChallengeSimulation	Workflow command	Re-runs or disputes assumptions, models, constraints, and interpretation.
L8-I05	Simulation Engine	SimulationInvalidated	Domain event	Marks results unfit and identifies dependent decision packages.
L9-I01	Decision Intelligence	OpenDecision	Command	Declares question, accountable authority, scope, objectives, options, stakeholders, and required evidence.
L9-I02	Decision Intelligence	DecisionPackageReady	Domain event	Publishes a complete reviewable package with uncertainty, alternatives, dissent, and provenance.
L9-I03	Decision Intelligence	ApproveDecision	Human command	Records authenticated approval, rationale, obligations, overrides, and commitment state.
L9-I04	Decision Intelligence	DecisionCommitted	Domain event	Activates commitments, execution handoff, monitoring conditions, and replay snapshot.
L9-I05	Decision Intelligence	DecisionReopened	Workflow event	Re-enters the decision lifecycle when evidence, assumptions, policy, or monitoring conditions change.
L10-I01	Executive Operating System	PublishBriefing	Command	Creates a live briefing bound to evidence, changes, objectives, decisions, and audience policy.
L10-I02	Executive Operating System	MaterialChangeRaised	Domain event	Routes change to accountable roles based on consequence, confidence, urgency, and attention policy.
L10-I03	Executive Operating System	ReviewConvened	Workflow event	Opens a governed review around a declared objective, decision, scenario, commitment, or outcome.
L10-I04	Executive Operating System	ExecutiveAction Requested	Human command	Requests analysis, scenario, simulation, decision, delegation, suppression, or follow-up.
L10-I05	Executive Operating System	AttentionPolicyChanged	Governance event	Versions materiality, escalation, suppression, and notification rules.

Common envelope

ENVELOPE FAMILY	REQUIRED FIELDS
Identity	tenant, domain, principal, workload or agent, actor assurance, delegation, session
Intent	purpose, action, object scope, expected product, side-effect class, approval context
Time	event, observation, valid, system, issue, deadline, expiry, and clock-quality context
Semantics	object type and version, schema and ontology version, truth state, locale, units
Trust	source and evidence references, transformation and model lineage, confidence, quality, coverage
Control	classification, policy revision, obligations, residency, key, retention, and export context
Operations	message ID, correlation, causation, idempotency, trace, sequence, partition, retry, checksum

D Layer Interaction Map

Canonical layers collaborate through objects and contracts. Direct database coupling, shared mutable state, or UI-mediated integration does not satisfy the architecture.

#	LAYER	PRIMARY UPSTREAM	PRIMARY DOWNSTREAM	CANONICAL INTERFACES
1	World Observation Layer	External and enterprise sources; source and policy authorities	Intelligence, Memory, Graph, Twins, Proactive Intelligence	RegisterSource, Acquire, ObservationRecorded, SourceHealthChanged, CorrectionReceived
2	Intelligence Layer	World Observation, ontology and model controls	Memory, Knowledge Graph, Twins, Prediction, Executive briefing	TransformEvidence, IntelligenceObjectProposed, ContradictionDetected, ReviewRequested, TransformationEvaluated
3	Enterprise Memory	All canonical layers and control planes	Retrieval, Graph, Twins, Decision Replay, Learning, Executive OS	CommitMemory, RetrieveContext, MemoryCorrected, RetentionActionDue, DeletionVerified
4	Knowledge Graph	Intelligence, Memory, ontology and stewardship	Twins, Prediction, Scenarios, Simulation, Decision, Executive OS	ResolveIdentity, CommitGraphRevision, GraphChanged, TraverseContext, OntologyChangeProposed
5	Digital Twins	Graph, Memory, enterprise systems, telemetry	Prediction, Scenario, Simulation, Decision, warnings	CreateTwinDefinition, ReconcileTwin, BranchTwin, TwinStateChanged, ValidateTwin
6	Prediction Engine	Graph, Memory, Twins, historical outcomes	Scenario, Simulation, Decision, warnings, Executive OS	RequestForecast, ForecastProduced, ForecastFitnessChanged, BacktestForecast, ForecastWithdrawn
7	Scenario Intelligence	Forecasts, Twins, Graph, Strategy Graph	Simulation, Decision, warning portfolio, Executive OS	CreateScenario, BranchScenario, ScenarioIndicatorChanged, ScenarioCoherenceFailed, ScenarioReviewed
8	Simulation Engine	Twins, scenarios, models, interventions	Decision Intelligence and Executive OS	SubmitExperiment, SimulationStarted, SimulationCompleted, ChallengeSimulation, SimulationInvalidated
9	Decision Intelligence	All intelligence and foresight layers	Executive OS, execution gateways, Decision Replay and Learning	OpenDecision, DecisionPackageReady, ApproveDecision, DecisionCommitted, DecisionReopened
10	Executive Operating System	Warnings, decisions, commitments, outcomes	Human action, governed workflows, review and learning	PublishBriefing, MaterialChangeRaised, ReviewConvened, ExecutiveActionRequested, AttentionPolicyChanged

Interaction constraints

- The Knowledge Graph supplies shared semantic identity but does not become a universal transactional database.
- Enterprise Memory preserves durable records; indexes, caches, vector representations, and graph projections are rebuildable views unless explicitly designated canonical.
- Digital Twins consume evidence and graph state through mappings and revisions; synthetic branches never write into observed state.
- Prediction, scenarios, and simulations publish versioned products; Decision Intelligence assembles but does not rewrite their uncertainty or lineage.
- The Executive Operating System organizes attention and action without creating a separate executive truth.

E Constitutional Traceability Matrix

Every founding invariant is anchored to architecture chapters. The full statement remains controlled by Volume 0; this matrix proves that Volume 3 neither omits nor silently reinterprets the constitutional baseline.

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 3 ANCHORS
C-001	Category integrity	The Eye shall be conceived, designed, sold, and operated as a Strategic Intelligence Operating System. It shall not be reduced to a chatbot, dashboard, BI tool, copilot, or reporting product.	Ch. 1, Ch. 54
C-002	Closed strategic loop	The product shall implement one continuous operating loop: Observe, Understand, Predict, Simulate, Decide, and Learn. No stage may be treated as a disconnected feature island.	Ch. 3
C-003	Product boundary	Interfaces such as chat, dashboards, reports, and APIs may exist, but they are access modes to the operating system rather than the product definition.	Ch. 2, Ch. 18
C-004	Human decision sovereignty	A named human authority shall make every final strategic or high-impact decision. The system may advise, prepare, and execute approved workflows, but it shall not silently assume decision rights.	Ch. 2, Ch. 17, Ch. 18, Ch. 23, Ch. 25, Ch. 37, Ch. 45, Ch. 52, Ch. 53
C-005	Explain every recommendation	Every material forecast, scenario, simulation result, risk, opportunity, and recommendation shall expose evidence, assumptions, uncertainty, lineage, alternatives, and accountable system versions.	Ch. 3, Ch. 7, Ch. 10, Ch. 14, Ch. 15, Ch. 16, Ch. 17, Ch. 21, Ch. 23, Ch. 24, Ch. 27, Ch. 29, Ch. 31, Ch. 34, Ch. 37, Ch. 50, Ch. 53
C-006	Enterprise operating standard	The common platform shall satisfy enterprise requirements for governance, security, scale, auditability, reliability, and lifecycle management from its first production architecture.	Ch. 1, Ch. 4, Ch. 8, Ch. 39, Ch. 54
C-007	Universal core, governed specialization	A single domain-neutral core shall serve all customer classes. Domain ontologies, models, workflows, agents, and policies may extend the core without forking its constitutional semantics.	Ch. 12, Ch. 52
C-008	Ten-layer architecture	The ten named architecture layers are the canonical product structure. Cross-cutting planes may support them but shall not replace, collapse, or rename them without constitutional amendment.	Ch. 1, Ch. 4, Ch. 54
C-009	Knowledge Graph centrality	The Knowledge Graph shall be the connected semantic heart of The Eye and the shared context substrate for memory, twins, reasoning, prediction, scenarios, simulations, and decisions.	Ch. 4, Ch. 12, Ch. 19, Ch. 31, Ch. 33
C-010	Truth-state separation	Observed evidence, extracted claims, inferred relationships, model outputs, human judgments, and synthetic scenario data shall remain explicitly distinguishable throughout the system.	Ch. 3, Ch. 7, Ch. 10, Ch. 12, Ch. 13, Ch. 20, Ch. 27, Ch. 31
C-011	Temporal truth	Material facts and relationships shall preserve event time, observation time, validity time, and system-record time so that past world states can be reconstructed without hindsight contamination.	Ch. 7, Ch. 8, Ch. 10, Ch. 11, Ch. 12, Ch. 13, Ch. 20, Ch. 27, Ch. 32, Ch. 33, Ch. 38, Ch. 51
C-012	End-to-end provenance	Every material intelligence object shall carry source, chain of custody, transformation history, ownership, rights, quality, and version lineage back to its evidence.	Ch. 7, Ch. 8, Ch. 9, Ch. 10, Ch. 12, Ch. 21, Ch. 30, Ch. 31, Ch. 32
C-013	Observation breadth	The World Observation Layer shall support the full canonical source universe and allow governed custom sources without architectural rework.	Ch. 9, Ch. 30
C-014	Freshness and quality contracts	Sources, derived data, twins, models, and intelligence products shall declare measurable freshness, completeness, quality, and expiry expectations.	Ch. 9, Ch. 30, Ch. 45, Ch. 46, Ch. 50
C-015	Durable institutional memory	The Eye shall preserve institutional context, decisions, assumptions, evidence, outcomes, and lessons as durable memory, subject to lawful retention, deletion, and customer policy.	Ch. 11, Ch. 38
C-016	Memory governance	Memory shall be permission-aware, tenant-isolated, versioned, retrievable, correctable, and protected from unverified model output becoming accepted organizational truth.	Ch. 7, Ch. 11, Ch. 27, Ch. 32
C-017	Strategy Graph	Objectives, hypotheses, decisions, initiatives, capabilities, resources, indicators, risks, opportunities, and outcomes shall be connected in a governed Strategy Graph anchored to the Knowledge Graph.	Ch. 12, Ch. 19
C-018	Grounded Digital Twins	Every Digital Twin shall be grounded in evidence, graph entities, temporal state, declared assumptions, and calibrated models; it shall never present synthetic state as observed fact.	Ch. 13, Ch. 16, Ch. 33, Ch. 36

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 3 ANCHORS
C-019	Twin versionability	Digital Twins shall support time travel, branching, comparison, replay, reconciliation, and versioned evolution across actual and simulated states.	Ch. 13, Ch. 16, Ch. 20, Ch. 33, Ch. 36
C-020	Canonical prediction horizons	The Prediction Engine shall support 30-day, 90-day, 180-day, 1-year, 3-year, and 5-year horizons with horizon-appropriate methods and uncertainty.	Ch. 14, Ch. 34
C-021	Probabilistic forecasting	Forecasts shall communicate distributions, confidence, calibration, drivers, sensitivity, and failure conditions rather than false precision or unsupported certainty.	Ch. 14, Ch. 34
C-022	Scenario plurality	Scenario Intelligence shall maintain multiple coherent futures, including baseline, upside, downside, disruption, and user-defined alternatives; no single forecast shall masquerade as destiny.	Ch. 15, Ch. 35
C-023	Reproducible simulation	Every simulation shall declare initial state, assumptions, interventions, model versions, constraints, random seeds where applicable, outputs, and sensitivity so results can be reproduced and challenged.	Ch. 13, Ch. 15, Ch. 16, Ch. 35, Ch. 36
C-024	Decision completeness	Decision Intelligence shall compare viable options against explicit objectives, constraints, risks, opportunities, trade-offs, second-order effects, reversibility, and information value.	Ch. 17, Ch. 19, Ch. 37
C-025	Decision Replay	The Eye shall preserve the state of knowledge available at decision time and support replay against later evidence and outcomes without rewriting the historical record.	Ch. 3, Ch. 7, Ch. 11, Ch. 17, Ch. 20, Ch. 25, Ch. 32, Ch. 37, Ch. 38, Ch. 51
C-026	Objective alignment	Recommendations shall be evaluated against declared objectives, stakeholder obligations, risk appetite, policy, and strategic constraints rather than an opaque global score.	Ch. 17, Ch. 19, Ch. 37
C-027	Executive Operating System	The executive surface shall organize strategic attention, decisions, scenarios, evidence, objectives, commitments, and outcomes. It shall not degrade into a passive dashboard collection.	Ch. 18
C-028	Bounded agent authority	Every agent shall have a declared purpose, tools, data scope, permissions, budgets, escalation rules, prohibited actions, evaluation suite, and accountable owner.	Ch. 6, Ch. 22, Ch. 23, Ch. 28
C-029	Supervised orchestration	Planner, Supervisor, and Workflow agents shall enforce policy, separation of duties, checkpoints, timeouts, retries, human gates, and complete execution traces across multi-agent work.	Ch. 23, Ch. 25, Ch. 31
C-030	Model pluralism	The platform shall support multiple models and reasoning approaches behind governed interfaces so customers retain control over cost, sovereignty, performance, and vendor dependence.	Ch. 10, Ch. 14, Ch. 24, Ch. 34
C-031	No silent self-modification	Continuous learning shall never permit unreviewed production models, agents, prompts, policies, ontologies, or decision logic to modify themselves silently.	Ch. 23, Ch. 24, Ch. 26, Ch. 38, Ch. 51
C-032	Proactive intelligence	The system shall surface material changes, weak signals, risks, and opportunities before users ask, while respecting relevance, authority, attention, and notification policy.	Ch. 15, Ch. 18, Ch. 29, Ch. 35
C-033	Early-warning discipline	Warnings shall be deduplicated, prioritized, evidence-backed, confidence-aware, time-bounded, and linked to affected objectives, assets, decisions, and response playbooks.	Ch. 15, Ch. 18, Ch. 29, Ch. 35
C-034	Decomposable Strategic Health Score	Any Strategic Health Score shall expose component measures, weights, evidence, confidence, trend, and sensitivity. The score shall never replace the underlying analysis.	Ch. 19, Ch. 29, Ch. 50
C-035	Trust as a system property	Trust, provenance, identity, authorization, policy, safety, and audit shall operate across every layer and intelligence object, not as a downstream compliance add-on.	Ch. 2, Ch. 4, Ch. 5, Ch. 6, Ch. 7, Ch. 8, Ch. 9, Ch. 12, Ch. 13, Ch. 16, Ch. 20, Ch. 21, Ch. 22, Ch. 24, Ch. 27, Ch. 28, Ch. 30, Ch. 33, Ch. 36, Ch. 44, Ch. 45, Ch. 46, Ch. 48, Ch. 49, Ch. 51, Ch. 53
C-036	Non-repudiable accountability	Material access, transformations, agent actions, approvals, decisions, overrides, releases, and exports shall produce tamper-evident audit records attributable to identities and versions.	Ch. 5, Ch. 17, Ch. 21, Ch. 22, Ch. 25, Ch. 32, Ch. 37, Ch. 44, Ch. 48, Ch. 49
C-037	Customer data control	Customers retain governance over their data, models, twins, policies, retention, residency, encryption, exports, and deletion within the chosen deployment and contractual boundary.	Ch. 6, Ch. 11, Ch. 22, Ch. 39, Ch. 40, Ch. 41, Ch. 42, Ch. 43, Ch. 44, Ch. 47
C-038	Open interoperability	The Eye shall integrate through governed APIs, events, connectors, common data formats, and export paths; core customer knowledge shall not be trapped behind proprietary-only access.	Ch. 8, Ch. 39, Ch. 41, Ch. 47, Ch. 51, Ch. 52

ID	INVARIANT	BINDING STATEMENT	PRIMARY VOLUME 3 ANCHORS
C-039	Zero-trust security	Identity, device, workload, agent, model, source, and data access shall be continuously authenticated, authorized, scoped, logged, and re-evaluated under least privilege.	Ch. 2, Ch. 5, Ch. 6, Ch. 9, Ch. 22, Ch. 30, Ch. 40, Ch. 41, Ch. 43, Ch. 44
C-040	Privacy and minimization	Collection and use shall be purpose-bound, proportionate, policy-enforced, and minimized, with support for consent, legal basis, access, correction, retention, and deletion obligations.	Ch. 5, Ch. 11, Ch. 22, Ch. 27, Ch. 40, Ch. 43, Ch. 44
C-041	Sovereignty by design	Data residency, key ownership, network isolation, local inference, model choice, administrative control, and disconnected operation shall be supported where customer policy requires them.	Ch. 5, Ch. 6, Ch. 24, Ch. 28, Ch. 39, Ch. 40, Ch. 41, Ch. 42, Ch. 43, Ch. 47
C-042	Deployment semantic parity	SaaS, Private Cloud, and On-Premise deployments shall preserve the same core semantics, governance model, APIs, artifacts, and intelligence lifecycle, with documented capability differences only where unavoidable.	Ch. 4, Ch. 39, Ch. 40, Ch. 41, Ch. 42, Ch. 43, Ch. 47, Ch. 49, Ch. 52, Ch. 53
C-043	Resilient operation	The platform shall fail safely, degrade intelligibly, recover predictably, preserve evidence and audit continuity, and expose when data, models, or dependencies are unavailable or stale.	Ch. 8, Ch. 9, Ch. 11, Ch. 13, Ch. 16, Ch. 25, Ch. 32, Ch. 36, Ch. 39, Ch. 40, Ch. 41, Ch. 42, Ch. 44, Ch. 45, Ch. 46, Ch. 47, Ch. 50, Ch. 53
C-044	Governed learning loop	Learning from outcomes, feedback, overrides, and new evidence shall pass through evaluation, approval, release, monitoring, rollback, and tenant-isolation controls.	Ch. 3, Ch. 26, Ch. 38
C-045	Continuous evaluation	Models, agents, forecasts, extractors, ontologies, simulations, and recommendations shall be evaluated for accuracy, calibration, robustness, drift, bias, safety, cost, and operational fitness.	Ch. 5, Ch. 10, Ch. 14, Ch. 16, Ch. 23, Ch. 24, Ch. 26, Ch. 31, Ch. 34, Ch. 36, Ch. 38, Ch. 46, Ch. 50, Ch. 51, Ch. 53
C-046	Governed marketplaces	Scenario and Agent Marketplace packages shall be signed, versioned, inspectable, permission-scoped, licensed, evaluated, sandboxed, monitored, revocable, and traceable to publishers.	Ch. 6, Ch. 15, Ch. 28, Ch. 35
C-047	Experience integrity	The product experience shall expose context, uncertainty, provenance, decision state, and consequence at the point of use; critical meaning shall not be hidden behind ornamental visualization.	Ch. 18, Ch. 29, Ch. 50, Ch. 52
C-048	Role and accessibility clarity	Experiences shall be accessible and role-appropriate for executives, analysts, operators, domain experts, administrators, and governance teams without creating separate truths.	Ch. 18, Ch. 50, Ch. 52
C-049	Operational quality contract	Reliability, performance, scale, freshness, accuracy, calibration, explainability, reproducibility, security, accessibility, and cost shall be measured as product behavior, not assumed as implementation detail.	Ch. 8, Ch. 14, Ch. 21, Ch. 25, Ch. 26, Ch. 29, Ch. 42, Ch. 45, Ch. 46, Ch. 49, Ch. 50, Ch. 53
C-050	Responsible-use boundary	The Eye shall not autonomously execute decisions that materially determine life, liberty, legal status, access to essential services, employment, credit, or equivalent high-impact rights and interests.	Ch. 2, Ch. 17, Ch. 26, Ch. 28, Ch. 37, Ch. 44, Ch. 53
C-051	Constitutional change control	Changes to mission, product category, human authority, principles, canonical layers, core semantics, or constitutional invariants require explicit amendment, impact analysis, ratification, and versioned publication.	Ch. 1, Ch. 48, Ch. 49, Ch. 54
C-052	Documentation inheritance	Volumes 1 through 10 shall inherit this constitution, use its canonical terms, trace material requirements to invariant IDs, and identify rather than conceal any unresolved conflict.	Ch. 1, Ch. 48, Ch. 54

F Architecture Glossary

These terms are canonical for Volumes 3–7. Domain extensions may add vocabulary but must map back to these meanings where the concepts overlap.

TERM	ARCHITECTURE DEFINITION
Agent Execution	A bounded, auditable run by an identified agent version under a declared purpose, plan, data scope, tools, budgets, supervision, and stop conditions.
Assessment	A versioned analytical judgment that separates evidence, method, uncertainty, dissent, and accountable author.
Canonical object	A governed intelligence object with stable identity, schema version, temporal semantics, truth state, provenance, policy, quality, and lifecycle.
Capability grant	A revocable authorization giving an agent or package a specific tool, data scope, model, network, side effect, budget, and time window.
Commitment	An approved institutional promise linked to a human decision, owner, resources, obligations, execution state, indicators, and closure criteria.
Conformance fixture	A portable test case that proves an implementation preserves canonical semantics and behavior across versions and deployment modes.
Correction	A versioned action that withdraws, replaces, adjudicates, merges, splits, or otherwise changes decision-active information without erasing history.
Customer Intelligence Domain	The primary logical isolation boundary for a customer's evidence, knowledge, models, agents, policies, decisions, and audit records.
Decision Package	A complete choice environment containing objectives, options, evidence, forecasts, scenarios, simulations, uncertainty, dissent, approvals, and monitoring.
Decision Replay	Reconstruction of the exact state of knowledge, authority, alternatives, rationale, commitments, and later outcomes of a decision.
Digital Twin	A governed, temporal, versioned representation of a declared real-world or enterprise boundary grounded in evidence, graph identity, models, and assumptions.
Domain event	An immutable statement that an authoritative state transition committed, carrying identity, time, causation, policy, and audit context.
Evidence Object	Preserved source material with source identity, chain of custody, checksum, classification, rights, transformations, quality, and coverage context.
Executive Operating System	The canonical layer governing strategic attention, responsibility, decisions, commitments, reviews, briefings, and outcomes.
Forecast Package	A horizon-specific probability product with distributions, drivers, calibration, sensitivity, failure conditions, and model lineage.
Graph revision	An atomic, validated set of temporal node, relationship, ontology, provenance, and stewardship changes.
Human gate	A durable workflow state requiring action by an authenticated human role with declared authority, information, options, and audit evidence.
Information boundary	The exact object, model, policy, and system versions available to a forecast, simulation, recommendation, or decision at a declared time.
Intelligence Cell	A scalable, isolated deployment and failure unit hosting a bounded set of customer or domain workloads.
Knowledge Graph	The central semantic substrate connecting canonical identities, events, relationships, evidence, time, provenance, and strategic objects.
Lesson Proposal	An evidence-backed proposed change to memory, models, agents, policy, workflow, ontology, or practice pending governance.
Material change	A change significant enough—by consequence, urgency, confidence, reversibility, or information value—to require accountable attention.
Model Gateway	The governed boundary that selects and invokes approved models under task, data, deployment, evaluation, cost, latency, and failure policy.

TERM	ARCHITECTURE DEFINITION
Observation Record	An immutable record that a source produced content or state at a particular observation time under a source contract.
Policy Decision	A recorded allow, deny, obligation, exception, or purpose determination bound to principal, object, action, environment, and time.
Provenance graph	The backward explanation and forward impact structure linking sources, evidence, transformations, objects, models, agents, approvals, and outcomes.
Quality contract	A measurable agreement defining a product or service indicator, population, target, window, failure behavior, recovery, and evidence.
Scenario Branch	A coherent, versioned plausible future with baseline, assumptions, actors, drivers, indicators, interventions, impacts, and review state.
Semantic parity	Preservation of object meaning, workflows, truth, time, provenance, authority, audit, and portability across SaaS, Private Cloud, and On-Premise.
Simulation Run	A reproducible experiment with exact twin, scenario, model, policy, environment, intervention, constraint, seed, output, and validation versions.
Source Contract	The governed definition of authority, rights, purpose, schema, latency, quality, authenticity, correction, and ownership for a source.
Strategic Health Score	A decomposable, governed roll-up of objective, risk, opportunity, capability, commitment, and outcome measures; never an opaque global judgment.
Strategic Memory	Durable institutional context spanning evidence, assessments, assumptions, decisions, commitments, outcomes, and lessons under memory governance.
Strategic Intelligence Operating System	A persistent system coordinating observation, understanding, prediction, simulation, human decision, and governed learning around a connected world and enterprise model.
Strategy Graph	The governed subgraph connecting objectives, assumptions, capabilities, risks, opportunities, initiatives, decisions, commitments, measures, and outcomes.
System time	When The Eye accepted or changed an object version, distinct from event, observation, and valid time.
Truth state	The explicit distinction among observed, claimed, inferred, assessed, decided, synthetic, withdrawn, and superseded information.
Valid time	The interval in the world during which an object or assertion applies.
Warning Product	A deduplicated proactive intelligence product stating change, evidence, uncertainty, consequence, response window, owner, and lifecycle.
Weak signal	A low-maturity but potentially material change with an explainable mechanism, evidence, uncertainty, strategic relevance, and monitoring path.

G Architecture Decision Baseline

These accepted decisions record the foundation of Volume 3. Detailed ADRs add context, alternatives, evidence, consequences, owners, and verification; they cannot supersede constitutional authority.

ID	DECISION	STATUS	BASELINE CONSEQUENCE	INHERITANCE
ADR-0001	Canonical ten-layer architecture	Accepted	The ten named layers are the product structure; control planes support but do not replace them.	C-008
ADR-0002	Knowledge Graph centrality	Accepted	Shared semantic identity and relationships are governed through the Knowledge Graph.	C-009
ADR-0003	Canonical intelligence object header	Accepted	All material objects carry identity, version, time, truth, provenance, control, quality, and operational correlation.	C-010–C-012
ADR-0004	Bitemporal system record	Accepted	Valid and system time are preserved alongside event and observation time.	C-011
ADR-0005	Immutable version history	Accepted	Corrections create new versions and propagation events; historical state is not overwritten.	C-011, C-025
ADR-0006	Customer Intelligence Domain	Accepted	Customer evidence, knowledge, models, agents, policies, decisions, and audit are isolated by a stable domain boundary.	C-037, C-039
ADR-0007	Human final authority	Accepted	Decision Intelligence may recommend and prepare; a named human makes final high-impact decisions.	C-004
ADR-0008	Durable workflow and human gates	Accepted	Consequential state transitions use durable workflows with explicit roles, evidence, approval, retry, and audit.	C-004, C-029
ADR-0009	Bounded agent execution	Accepted	Agents operate through capability grants, supervision, budgets, evaluation, and stop conditions.	C-028, C-029
ADR-0010	Model gateway and pluralism	Accepted	Models remain replaceable, governed dependencies resolved by task and deployment policy.	C-030
ADR-0011	Digital Twin state separation	Accepted	Observed, estimated, forecast, scenario, and simulated twin states remain distinct and versioned.	C-018, C-019
ADR-0012	Probabilistic forecast package	Accepted	Forecast products carry distributions, calibration, sensitivity, lineage, and failure conditions.	C-020, C-021
ADR-0013	Scenario portfolio	Accepted	Multiple coherent branches remain visible and monitored; no branch becomes destiny.	C-022
ADR-0014	Reproducible simulation manifest	Accepted	Simulation binds exact inputs, versions, constraints, environment, seeds, outputs, and validation.	C-023
ADR-0015	Decision package completeness	Accepted	Objectives, options, evidence, uncertainty, trade-offs, dissent, authority, and monitoring form one package.	C-024–C-026
ADR-0016	Deployment semantic parity	Accepted	SaaS, Private Cloud, and On-Premise implement one product contract.	C-042
ADR-0017	Sovereignty as policy dimensions	Accepted	Data, keys, models, administration, telemetry, supply, support, and continuity are independently governed.	C-041
ADR-0018	Safe degradation hierarchy	Accepted	Failure may reduce capability but cannot fabricate currency, confidence, authorization, or approval.	C-043, C-049
ADR-0019	Governed marketplace packages	Accepted	Packages are signed, declared, sandboxed, evaluated, capability-scoped, monitored, and revocable.	C-046
ADR-0020	Governed learning release	Accepted	Lessons become behavior only through evaluation, approval, deployment, monitoring, and rollback.	C-031, C-044, C-045

Baseline closure

Volume 3 closes when all downstream architecture and engineering work can identify its controlling layer, component, object, interface, runtime flow, trust boundary, quality contract, deployment invariant, and constitutional reference. The architecture remains frozen through that inheritance chain until an authorized change is ratified and propagated across the documentation set.

END STATE

The Eye is defined here as one coherent Strategic Intelligence Operating System: continuously observing, constructing governed understanding, preserving institutional memory, connecting the world through knowledge and twins, forecasting and simulating possible futures, preparing explainable decisions for humans, governing commitments, and learning from outcomes.

CONSTITUTIONAL INHERITANCE / C-001 · C-002 · C-004 · C-005 · C-051 · C-052